QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

trezor-vendored-crypto @ 72489948b3c4 · Trezor (vendored primitives) — public repository (passive review)

Organization	QSI
Report date	2026-10-29 05:35 UTC
Report version	1
Audit run	2479351b-857a-4bec-888f-076ff6716d7d · Document audit · 2026-10-29 05:35 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`0946734472e0712a577bfef0a07be6b9e19572e9b4966a66229e52b4f0abe051`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

4	0	0	4	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	4
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-06405	Medium	Quantum-vulnerable ECDSA: transaction and message signing (24 file(s)) confirmed by the analyst	41.8
QSI-2026-06406	Medium	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (4 file(s)) confirmed by the analyst	41.8
QSI-2026-06404	Medium	Quantum-vulnerable Ed25519: role not determined (5 file(s)) confirmed by the analyst	40.2
QSI-2026-06407	Medium	Quantum-vulnerable ECDH: role not determined (1 file(s)) confirmed by the analyst	40.2

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Analyst review: 4 confirmed, 0 not applicable, 0 dismissed as false positives.

Post-quantum maturity: Level 1 of 4 — post-quantum code present, not wired. Next step: Wire the existing post-quantum primitive into a critical path (key wrapping or the channel) behind a hybrid construction, and test interoperability.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/trezor/trezor-firmware @ 72489948b3c44a37598de9da658be28e4fabb4ae — evidence links open the lines at this commit.

Supplied inputs

519 files, 5085520 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	519	vendor/mldsa-native/.github/actions/cbmc/report.py: 297 lines scanned with 16 rule patterns and 123 registry patterns; vendor/mldsa-native/dev/aarch64_clean/meta.h: 247 lines scanned with 16 rule patterns and 123 registry patterns; vendor/mldsa-native/dev/aarch64_clean/src/aarch64_zetas.c: 231 lines scanned with 16 rule patterns and 123 registry patterns

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 519 (input files: C)

Architecture overview

Static review of the source tree trezor-vendored-crypto at 72489948b3c44a37598de9da658be28e4fabb4ae. 519 files analysed; of 1159 files, 174 are test, fuzzing, benchmark or example code and were left out, 466 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. 519 files come from directories a .gitmodules pins and are read as shipped source rather than skipped as a dependency cache. No requests were sent to the owner's systems.

Overall security posture

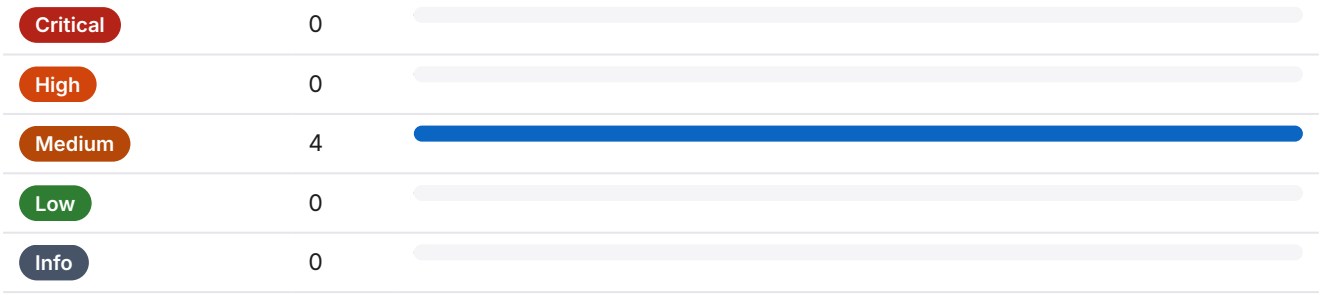
100
Indicative posture index (0–100)

26.8%
Quantum-vulnerable share of inventory

142
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-06405	Quantum-vulnerable ECDSA: transaction and message signing (24 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06406	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (4 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06404	Quantum-vulnerable Ed25519: role not determined (5 file(s))	Long-term post-quantum risk	Medium	Medium
QSI-2026-06407	Quantum-vulnerable ECDH: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium

Detailed findings

QSI-2026-06405 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (24 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-29): confirmed by the analyst.** Correct: 24 files of vendor/secp256k1-zkp, Blockstream's fork of libsecp256k1, which is what a Trezor actually signs Bitcoin transactions with. The main audit's 73 Shor-breakable rows count the call sites in core/embed/sec; these are the implementation under them. The two should be read together and not added together — it is one dependency measured at two levels, which is why this bundle is not folded into the portfolio's per-codebase counts.

82 reference(s) to ECDSA across 24 file(s), e.g. vendor/secp256k1-zkp/contrib/lax_der_parsing.c:11. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	vendor/secp256k1-zkp/contrib/lax_der_parsing.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:35 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open vendor/secp256k1-zkp/contrib/lax_der_parsing.c:11 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.c · line 11 · SHA-256 9197354c850cb53b… · File 4da2ecdc0a3b9dbb…

```
   10 |
   11 | int ecdsa_signature_parse_der_lax(const secp256k1_context* ctx, secp256k1_ecdsa_signature*
sig, const unsigned char *input, size_t inputlen) {
   12 |     size_t rpos, rlen, spos, slen;
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.c · line 19 · SHA-256 234aeaaf6b28e1d2… · File 4da2ecdc0a3b9dbb…

```
   18 |     /* Hack to initialize sig with a correctly-parsed but invalid signature. */
   19 |     secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
   20 |
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.c · line 140 · SHA-256 75ad5d0cba25ca64… · File 4da2ecdc0a3b9dbb…

```
139 |     if (!overflow) {
140 |         overflow = !secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
141 |     }
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.h · line 57 · SHA-256 25af7e242d92af0d… · File 7f3e8ba009f19a5f…

```
 56 |    files (such as tests.c) without the need to set an explicit -I flag,
 57 |    which would be necessary to locate secp256k1.h. */
 58 | #ifndef SECP256K1_H
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.h · line 59 · SHA-256 4c35a81e99a7eaf5… · File 7f3e8ba009f19a5f…

```
 58 | #ifndef SECP256K1_H
 59 | #include <secp256k1.h>
 60 | #endif
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_parsing.h · line 88 · SHA-256 85efcade66268dd4… · File 7f3e8ba009f19a5f…

```
 87 |     const secp256k1_context* ctx,
 88 |     secp256k1_ecdsa_signature* sig,
 89 |     const unsigned char *input,
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h · line 34 · SHA-256 163cac24f6b60e79… · File 3420812d623c0f2f…

```
 33 |    files (such as tests.c) without the need to set an explicit -I flag,
 34 |    which would be necessary to locate secp256k1.h. */
 35 | #ifndef SECP256K1_H
```

file_excerpt · vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h · line 36 · SHA-256 2741d1869779ddac… · File 3420812d623c0f2f…

```
 35 | #ifndef SECP256K1_H
 36 | #include <secp256k1.h>
 37 | #endif
```

file_excerpt · vendor/secp256k1-zkp/include/secp256k1.h · line 74 · SHA-256 712d2259ac1032ad… · File 328da0a2e0433e9c…

```
 73 |  */
 74 | typedef struct secp256k1_ecdsa_signature {
 75 |     unsigned char data[64];
```

file_excerpt · vendor/secp256k1-zkp/include/secp256k1.h · line 76 · SHA-256 2ccd260bdbfd5dbc… · File 328da0a2e0433e9c…

```
 75 |     unsigned char data[64];
 76 | } secp256k1_ecdsa_signature;
 77 |
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06406 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (4 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-29): confirmed by the analyst.** Correct, four files, the schnorrsig module of the same vendored library: Taproot spends.

25 reference(s) to Schnorr (BIP-340) across 4 file(s), e.g. vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h:82. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Schnorr (BIP-340):transaction_signing
Data source	vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:35 UTC
Priority score	41.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h:82 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 82 · SHA-256 6aae3b20f8f8b503… · File 8185d6c6ea3234ff…

```
  */
typedef struct secp256k1_schnorrsig_extraparams {
    unsigned char magic[4];
```

file_excerpt · vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 86 · SHA-256 43d5f6045ac4e6b2… · File 8185d6c6ea3234ff…

```
   85 |     void *ndata;
   86 | } secp256k1_schnorrsig_extraparams;
   87 |
```

file_excerpt · vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h · line 119 · SHA-256 1e555d327421285c… · File 8185d6c6ea3234ff…

```
  118 |  */
  119 | SECP256K1_API int secp256k1_schnorrsig_sign32(
  120 |     const secp256k1_context *ctx,
```

file_excerpt · vendor/secp256k1-zkp/src/modules/musig/session_impl.h · line 626 · SHA-256 a9274c7d82071c2c… · File 9dcc4c823bd1bb9c…

```
  625 |     secp256k1_musig_nonce_process_internal(&session_i.fin_nonce_parity, fin_nonce,
&session_i.noncecoef, aggnonce_pts, agg_pk32, msg32);
  626 |     secp256k1_schnorrsig_challenge(&session_i.challenge, fin_nonce, msg32, 32, agg_pk32);
  627 |
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 33 · SHA-256 fc2c5559f32fc9bd… · File 1e9f3b239e1fac3f…

```
   32 |         msg[1] = i >> 8;
   33 |         CHECK(secp256k1_schnorrsig_sign_custom(data->ctx, sig, msg, MSGLEN, data-
>keypairs[i], NULL));
   34 |     }
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 44 · SHA-256 4f2e9b7b2529431d… · File 1e9f3b239e1fac3f…

```
   43 |         CHECK(secp256k1_xonly_pubkey_parse(data->ctx, &pk, data->pk[i]) == 1);
   44 |         CHECK(secp256k1_schnorrsig_verify(data->ctx, data->sigs[i], data->msgs[i], MSGLEN,
&pk));
   45 |     }
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h · line 80 · SHA-256 0ca8c9b515200145… · File 1e9f3b239e1fac3f…

```
   79 |         CHECK(secp256k1_keypair_create(data.ctx, keypair, sk));
   80 |         CHECK(secp256k1_schnorrsig_sign_custom(data.ctx, sig, msg, MSGLEN, keypair, NULL));
   81 |         CHECK(secp256k1_keypair_xonly_pub(data.ctx, &pk, NULL, keypair));
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h · line 106 · SHA-256 010e6d7bb2007942… · File d627be29e9a7e1e3…

```
  105 |  * SHA256 to SHA256("BIP0340/challenge")||SHA256("BIP0340/challenge"). */
  106 | static void secp256k1_schnorrsig_sha256_tagged(secp256k1_sha256 *sha) {
  107 |     secp256k1_sha256_initialize(sha);
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h · line 119 · SHA-256 4a22e3f79b439b47… · File d627be29e9a7e1e3…

```
  118 |
  119 | static void secp256k1_schnorrsig_challenge(secp256k1_scalar* e, const unsigned char *r32,
const unsigned char *msg, size_t msglen, const unsigned char *pubkey32)
  120 | {
```

file_excerpt · vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h · line 125 · SHA-256 521a9c533807ec41… · File d627be29e9a7e1e3…

```
124 |     /* tagged hash(r.x, pk.x, msg) */
125 |     secp256k1_schnorrsig_sha256_tagged(&sha);
126 |     secp256k1_sha256_write(&sha, r32, 32);
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06404 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (5 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-29): confirmed by the analyst.** Correct and directly relevant to the migration story rather than to spending. Ed25519 is what the device's boot header is signed with today — core/embed/sec/image/boot_header.c:81 in the main audit — and these five rows are its implementation. This is the primitive SLH-DSA is being added alongside, so the pair Ed25519-plus-SLH-DSA in one header is the whole point: the root of trust is mid-migration, carrying both. Firmware signatures are forward-only exposure, but the consequence of a forgery is arbitrary code on a device holding keys, which is why a signer's boot path is where post-quantum signatures matter first even though the spending path is the louder number.

6 reference(s) to Ed25519 across 5 file(s), e.g. vendor/mldsa-native/mldsa/mldsa_native.c:212. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	vendor/mldsa-native/mldsa/mldsa_native.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:35 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open vendor/mldsa-native/mldsa/mldsa_native.c:212 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · vendor/mldsa-native/mldsa/mldsa_native.c · line 212 · SHA-256 d866878f98e2f9d8… · File cea0f7cd3c72998b…

```
211 | #undef crypto_sign
212 | #undef crypto_sign_keypair
213 | #undef crypto_sign_open
```

file_excerpt · vendor/mldsa-native/mldsa/mldsa_native.h · line 866 · SHA-256 cdc8f37b350eb974… · File 01ae99fa0e65a17f…

```
865 |
866 | #define crypto_sign_keypair MLD_API_NAMESPACE(keypair)
867 | #define crypto_sign_signature MLD_API_NAMESPACE(signature)
```

file_excerpt · vendor/sphincsplus/ref/PQCgenKAT_sign.c · line 118 · SHA-256 86be2a14cdae0837… · File 793da075036e9068…

```
117 |         // Generate the public/private keypair
118 |         if ( (ret_val = crypto_sign_keypair(pk, sk)) != 0) {
119 |             printf("crypto_sign_keypair returned <%d>\n", ret_val);
```

file_excerpt · vendor/sphincsplus/ref/PQCgenKAT_sign.c · line 119 · SHA-256 26d224d32ba13de8… · File 793da075036e9068…

```
118 |         if ( (ret_val = crypto_sign_keypair(pk, sk)) != 0) {
119 |             printf("crypto_sign_keypair returned <%d>\n", ret_val);
120 |             return KAT_CRYPTO_FAILURE;
```

file_excerpt · vendor/sphincsplus/ref/api.h · line 49 · SHA-256 639ca6453aba334e… · File a0c7abd648a4a695…

```
 48 |  */
 49 | int crypto_sign_keypair(unsigned char *pk, unsigned char *sk);
 50 |
```

file_excerpt · vendor/sphincsplus/ref/sign.c · line 83 · SHA-256 689d2d643099c5e9… · File 4015a353fa2b0433…

```
 82 |  */
 83 | int crypto_sign_keypair(unsigned char *pk, unsigned char *sk)
 84 | {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06407 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium

> **Analyst review (QSI review, 2026-10-29): confirmed by the analyst.** One row, and the only retroactive one here: key agreement in the vendored secp256k1-zkp. Where a shared secret derived through it protects anything stored or recorded, a copy taken today becomes readable once a CRQC exists. Unlike the signature rows it can be hardened without a consensus change, by wrapping the agreed secret under a hybrid KEM.

4 reference(s) to ECDH across 1 file(s), e.g. vendor/secp256k1-zkp/src/bench.c:24. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDH:general
Data source	vendor/secp256k1-zkp/src/bench.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-29 05:35 UTC
Priority score	40.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open vendor/secp256k1-zkp/src/bench.c:24 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · vendor/secp256k1-zkp/src/bench.c · line 24 · SHA-256 6ec91ca104f465f4… · File 08b5b70457175cfd…

```
  23 | #ifdef ENABLE_MODULE_ECDH
  24 |     printf("    - ECDH key exchange (optional module)\n");
  25 | #endif
```

file_excerpt · vendor/secp256k1-zkp/src/bench.c · line 54 · SHA-256 b21ecd8d53157d4f… · File 08b5b70457175cfd…

```
53 | #ifdef ENABLE_MODULE_ECDH
54 |     printf("    ecdh              : ECDH key exchange algorithm\n");
55 | #endif
```

file_excerpt · vendor/secp256k1-zkp/src/bench.c · line 68 · SHA-256 ae7392892e7a0840… · File 08b5b70457175cfd…

```
67 |     printf("    ellswift_keygen   : ElligatorSwift key generation\n");
68 |     printf("    ellswift_ecdh     : ECDH on ElligatorSwift keys\n");
69 | #endif
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

Evidence

ID	Type	Data source	SHA-256
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.c (line 11)	9197354c850cb53bbc475b348c0f97bb3f550706fcced8bf483336d6
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.c (line 19)	234aeaaf6b28e1d2ff43854e60a3119e3c3f55229e18841e2dc7664(
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.c (line 140)	75ad5d0cba25ca64adb6d971da937879704b4abc8845c17dc1da9fa1
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.h (line 57)	25af7e242d92af0d3c46ae8d8a48030e247c4bfdeba65068030e64eb
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.h (line 59)	4c35a81e99a7eaf5ea7315346f3b8cacb0832f7c4a3b1a3d003ff0bc
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_parsing.h (line 88)	85efcade66268dd4d0b42e8f31b634ea4cd93d9d55cdc2e036715108
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h (line 34)	163cac24f6b60e798317e2383fb603da718e728035262dc9759d7cf3
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h (line 36)	2741d1869779ddac4f151ac1740c338ba299b2bed7b98a38e6042acb
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/include/secp256k1.h (line 74)	712d2259ac1032ad406ddde4403d5a72e5739b8c7e004ad0b2293d26
QSI-2026-06405	file_excerpt	vendor/secp256k1-zkp/include/secp256k1.h (line 76)	2ccd260bdbfd5dbcb2816a89d4808a078f7cf59fef64a70244b731ba
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 82)	6aae3b20f8f8b503597acb97fe491246b6777b5a9660b8a41542605c
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 86)	43d5f6045ac4e6b23daf11fd6c10022dca16ca76afd775c3ee898ddb

QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h (line 119)	`1e555d327421285c8a81345e58876f5f2f017571adb2f2168aafc7c4`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/musig/session_impl.h (line 626)	`a9274c7d82071c2c191759a1ad6acc91c4341fab17e9bfb18c95c5a2`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 33)	`fc2c5559f32fc9bd9a85efef59a46eaf722b1b8e2745b837696cdf6a`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 44)	`4f2e9b7b2529431d8190d606bbe0ce49659d6e5a1e0410625d931eac`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h (line 80)	`0ca8c9b51520014576b4c622e33014e8b93cc7c4bd0bc826e9cd4563`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h (line 106)	`010e6d7bb200794264a2bce239dc7cc1cf2fb59c2ce7595deb9f5d4c`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h (line 119)	`4a22e3f79b439b475e0b65863dad24c6233db490ff1eedb9bb5d1a4c`
QSI-2026-06406	file_excerpt	vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h (line 125)	`521a9c533807ec4184854d2f7d1ebe0afbbe1f0b7ec13d0c705a0ab7`
QSI-2026-06404	file_excerpt	vendor/mldsa-native/mldsa/mldsa_native.c (line 212)	`d866878f98e2f9d8188f28872711433022de6c18d69a1561655c918b`
QSI-2026-06404	file_excerpt	vendor/mldsa-native/mldsa/mldsa_native.h (line 866)	`cdc8f37b350eb9742546761cc89f9ecfc27e71b175d4e86246d3494c`
QSI-2026-06404	file_excerpt	vendor/sphincsplus/ref/PQCgenKAT_sign.c (line 118)	`86be2a14cdae0837afaa6df56e5c682e5cd4574050a77bd7b777631c`
QSI-2026-06404	file_excerpt	vendor/sphincsplus/ref/PQCgenKAT_sign.c (line 119)	`26d224d32ba13de87a6b88d4993e9684549d45fae1b6992ac390883a`
QSI-2026-06404	file_excerpt	vendor/sphincsplus/ref/api.h (line 49)	`639ca6453aba334ed81ce97556bd5c8c134495bf530bc8e7b87b50a8`
QSI-2026-06404	file_excerpt	vendor/sphincsplus/ref/sign.c (line 83)	`689d2d643099c5e9f712db4d28751748d7fff094f1301c64ccaeae54`
QSI-2026-06407	file_excerpt	vendor/secp256k1-zkp/src/bench.c (line 24)	`6ec91ca104f465f4b861193bee0ed1f6d91c4074fb5085511ca06c7c`
QSI-2026-06407	file_excerpt	vendor/secp256k1-zkp/src/bench.c (line 54)	`b21ecd8d53157d4f920477f57713fa34b7fe6379df61daa64f27e4a7`
QSI-2026-06407	file_excerpt	vendor/secp256k1-zkp/src/bench.c (line 68)	`ae7392892e7a0840c3596f3442ca87927d89cc694ec91267d2a73193`

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 1 of 4 — post-quantum code present, not wired

- Present, not wired: ML-DSA — vendor/mldsa-native/integration/liboqs/config_aarch64.h
- Present, not wired: ML-DSA — vendor/mldsa-native/integration/liboqs/config_c.h
- Present, not wired: ML-DSA — vendor/mldsa-native/integration/liboqs/config_x86_64.h
- Present, not wired: ML-DSA — vendor/mldsa-native/mldsa/mldsa_native.c
- Present, not wired: ML-DSA — vendor/mldsa-native/mldsa/mldsa_native.h

Next step: Wire the existing post-quantum primitive into a critical path (key wrapping or the channel) behind a hybrid construction, and test interoperability.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (24), Schnorr (BIP-340) (4)	QSI-2026-06405, QSI-2026-06406	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
role not determined	Ed25519 (5), ECDH (1)	QSI-2026-06404, QSI-2026-06407	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	1
Grover: acceptable to NIST, below ECCG/ANSSI guidance	1
post-quantum	102
vulnerable (Shor)	38

Detailed findings: QSI-2026-06405, QSI-2026-06406, QSI-2026-06404, QSI-2026-06407

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDH (1), ECDSA (28), Ed25519 (5), Schnorr (BIP-340) (4)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	AES-256 (1), SHA-2 (1), SLH-DSA (94)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (1)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.
Quantum lattice sieving Q1: quantum computation offline, classical access to the victim	included in standard parameters	ML-DSA (8)	No change for ML-KEM or ML-DSA at the standard parameter sets. A 2024 claim of a polynomial-time quantum LWE algorithm was withdrawn after a bug was found (see claims).

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-06405 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06406 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06404 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06407 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06405 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06406 · P4 · 90 days

root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-06404 · P4 · 90 days

root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06407 · P4 · 90 days

root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-06405	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06406	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06404	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06407	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-06405	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06406	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06404	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06407	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-06405	Re-run the inventory after refactoring.
QSI-2026-06406	Re-run the inventory after refactoring.
QSI-2026-06404	Re-run the inventory after refactoring.
QSI-2026-06407	Re-run the inventory after refactoring.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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2. A. Chailloux, J. Loyer — Lattice sieving via quantum random walks (ASIACRYPT 2021; IACR ePrint 2021/570) — IACR ePrint. https://eprint.iacr.org/2021/570 (peer-reviewed article: 2^{0.2570d + o(d)}, previously 2^{0.2653d + o(d)} (Laarhoven))
3. A. Chailloux, M. Naya-Plasencia, A. Schrottenloher — An Efficient Quantum Collision Search Algorithm and Implications on Symmetric Cryptography (ASIACRYPT 2017; IACR ePrint 2017/847) — IACR ePrint. https://eprint.iacr.org/2017/847 (peer-reviewed article: Õ(2^{2n/5}) time, O(n) qubits, Õ(2^{n/5}) classical memory)
4. C. Chevignard, P.-A. Fouque, A. Schrottenloher — Reducing the Number of Qubits in Quantum Discrete Logarithms on Elliptic Curves (IACR ePrint 2026/280) — IACR ePrint. https://eprint.iacr.org/2026/280 (revision of a EUROCRYPT 2026 paper: 3.12n + o(n) qubits; for n = 256, 1193 qubits and 22 runs of 2^38.98 Toffoli gates)
5. C. Zalka — Grover's quantum searching algorithm is optimal (Phys. Rev. A 60, 2746, 1999; arXiv:quant-ph/9711070) — APS. https://doi.org/10.1103/PhysRevA.60.2746 (peer-reviewed article: optimal for any success probability; parallel search gains no more than splitting the space)
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7. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
8. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
9. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
10. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
11. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
12. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
13. G. Brassard, P. Høyer, A. Tapp — Quantum Algorithm for the Collision Problem (LATIN '98; arXiv:quant-ph/9705002) — Springer / arXiv. https://arxiv.org/abs/quant-ph/9705002 (peer-reviewed article: collisions of r-to-one functions in O((N/r)^{1/3}) expected evaluations)
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15. L. K. Grover — A fast quantum mechanical algorithm for database search (STOC 1996) — ACM. https://doi.org/10.1145/237814.237866 (peer-reviewed article)
16. M. Amy et al. — Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3 (SAC 2016; arXiv:1603.09383) — Springer / arXiv. https://arxiv.org/abs/1603.09383 (peer-reviewed article: SHA-256 about 2^153.8 surface-code cycles on 2^12.6 logical qubits; SHA3-256 about 2^146.5 cycles on 2^20)
17. M. Cain et al. — Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) — arXiv. https://arxiv.org/abs/2603.28627 (preprint (2026-03-30): P-256 in a few days with 26,000 physical neutral-atom qubits; RSA-2048 one to two orders of magnitude longer)
18. M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018) — IEEE. https://doi.org/10.1109/MSP.2018.3761723 (peer-reviewed article; source of the x + y > z migration inequality)
19. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
20. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/ipd (initial public draft (November 2024); proposes deprecating quantum-vulnerable public-key algorithms after 2030 and disallowing them after 2035 — dates may change in the final version)
21. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/

Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)

22. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
23. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)
24. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
AES-256	AES-256	reduced margin (Grover), approved	acceptable	1	1	vendor/sphincsplus/ref/rng.c:125
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	1	1	vendor/sphincsplus/vectors.py:59
ML-DSA		post-quantum	acceptable	8	8	vendor/mldsa-native/integration/liboqs/config_aarch64.h:315; vendor/mldsa-native/integration/liboqs/config_c.h:319; vendor/mldsa-native/integration/liboqs/config_x86_64.h:317
SLH-DSA		post-quantum	acceptable	94	94	vendor/sphincsplus/haraka-aesni/context.h:10; vendor/sphincsplus/haraka-aesni/haraka.c:117; vendor/sphincsplus/haraka-aesni/harakax4.h:19
ECDH		vulnerable (Shor)	acceptable	1	1	vendor/secp256k1-zkp/src/bench.c:24
ECDSA		vulnerable (Shor)	acceptable	7	7	vendor/secp256k1-zkp/contrib/lax_der_parsing.c:11; vendor/secp256k1-zkp/contrib/lax_der_parsing.h:88; vendor/secp256k1-zkp/include/secp256k1.h:74
ECDSA	secp256k1	vulnerable (Shor)	acceptable	21	21	vendor/secp256k1-zkp/contrib/lax_der_parsing.h:57; vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h:34; vendor/secp256k1-zkp/include/secp256k1_ecdh.h:4
Ed25519	Ed25519	vulnerable (Shor)	acceptable	5	5	vendor/mldsa-native/mldsa/mldsa_native.c:212; vendor/mldsa-native/mldsa/mldsa_native.h:866; vendor/sphincsplus/ref/PQCgenKAT_sign.c:118
Schnorr (BIP-340)		vulnerable (Shor)	acceptable	4	4	vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h:82; vendor/secp256k1-zkp/src/modules/musig/session_impl.h:626; vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h:33

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 0946734472e0712a577bfef0a07be6b9e19572e9b4966a66229e52b4f0abe051

C. Input files

File	Size	SHA-256
vendor/mldsa-native/.github/actions/cbmc/report.py	8847	6fa18f2c8481cc906f39b461fcf770b78b81af81d5e04183d0daf
vendor/mldsa-native/dev/aarch64_clean/meta.h	9372	86458b409219c5b521acd1c8e780a7dd5b680ab1607f548861cbf
vendor/mldsa-native/dev/aarch64_clean/src/aarch64_zetas.c	14368	87e22e4dd92e86f655870785fc8a6f40be187b543c0466920092f
vendor/mldsa-native/dev/aarch64_clean/src/arith_native_aarch64.h	6497	406b41271328f5f5488b25a2dd061787c2d73bc4e1cc1e0b13ce7
vendor/mldsa-native/dev/aarch64_clean/src/polyz_unpack_table.c	1417	0961cb66fd4951a3b01a243cf6886949d497b7ed19fefc7264d45
vendor/mldsa-native/dev/aarch64_clean/src/rej_uniform_eta_table.c	25597	9bd3425816dcc0c21115f6955cc8830c4cdc3c7bebb1e1d6ed76e
vendor/mldsa-native/dev/aarch64_clean/src/rej_uniform_table.c	2295	3308702a2208ea746e77c02792fa47ccfd0ea7ed7fc439e7cbcfd
vendor/mldsa-native/dev/aarch64_opt/meta.h	9372	86458b409219c5b521acd1c8e780a7dd5b680ab1607f548861cbf
vendor/mldsa-native/dev/aarch64_opt/src/aarch64_zetas.c	14368	87e22e4dd92e86f655870785fc8a6f40be187b543c0466920092f
vendor/mldsa-native/dev/aarch64_opt/src/arith_native_aarch64.h	6497	406b41271328f5f5488b25a2dd061787c2d73bc4e1cc1e0b13ce7
vendor/mldsa-native/dev/aarch64_opt/src/polyz_unpack_table.c	1417	0961cb66fd4951a3b01a243cf6886949d497b7ed19fefc7264d45
vendor/mldsa-native/dev/aarch64_opt/src/rej_uniform_eta_table.c	25597	9bd3425816dcc0c21115f6955cc8830c4cdc3c7bebb1e1d6ed76e
vendor/mldsa-native/dev/aarch64_opt/src/rej_uniform_table.c	2295	3308702a2208ea746e77c02792fa47ccfd0ea7ed7fc439e7cbcfd
vendor/mldsa-native/dev/fips202/aarch64/auto.h	1988	b47134bffe6ce1469e0c94c773d7fc03f317e9882917347de0f56
vendor/mldsa-native/dev/fips202/aarch64/src/fips202_native_aarch64.h	2478	184ac5106fe5e4501fb10bb1e0db4c5b55691c0d7647db00c9eb7
vendor/mldsa-native/dev/fips202/aarch64/src/keccakf1600_round_constants.c	1996	c9351871c008c57156ea5b5192e39ffbef7dc617bd944c640ef8b
vendor/mldsa-native/dev/fips202/aarch64/x1_scalar.h	757	6f6204bf6b4a6bf83c3e8a8c2dd926832269fdd5d0d5a851af982
vendor/mldsa-native/dev/fips202/aarch64/x1_v84a.h	956	9ba4f0efbe1c57790fa2bfdad454e435878610adb92738e454f85
vendor/mldsa-native/dev/fips202/aarch64/x2_v84a.h	1047	9f2db3cabcd572733ca96019640200dac1d23729395c3d9f93377

vendor/mldsa-native/dev/fips202/aarch64/x4_v8a_scalar.h	835	fd36b58eab09c739f064d062437dc849345f255e7119b71a3e653
vendor/mldsa-native/dev/fips202/aarch64/x4_v8a_v84a_scalar.h	1032	a63e538011f4176b0e083f42cc35ef7c79ce1724fc21fb756c130
vendor/mldsa-native/dev/fips202/armv81m/mve.h	813	0e2257310cfa60a98474c1d9201d7306f42ed5846406c37e4d8f1
vendor/mldsa-native/dev/fips202/armv81m/src/fips202_native_armv81m.h	794	19f6ab96c9005836e0d2b945ac2bcb15f1b976fa4d19fd05869de
vendor/mldsa-native/dev/fips202/armv81m/src/keccak_f1600_x4_mve.c	4687	03dd17852b4827d662ff622de4070b392ca5c590ef36de0cad9d6
vendor/mldsa-native/dev/fips202/armv81m/src/keccakf1600_round_constants.c	1729	9b8bf66e1bbf69d8ada688d0ebb029a4b57d1053ce3b0f6767240
vendor/mldsa-native/dev/x86_64/meta.h	10542	5a629bba590fc8b7f8ade17e07a7e28ad93cf2e7007d8bf91fcbf
vendor/mldsa-native/dev/x86_64/src/arith_native_x86_64.h	5074	e52ef79ddefd09e1bf941a2794435393e8e2cd8c9e1916504342b
vendor/mldsa-native/dev/x86_64/src/consts.c	9085	25ab5b8f9def84990da5cd747f2ff2298c684c6c6fc2441ed79c4
vendor/mldsa-native/dev/x86_64/src/consts.h	806	9f19c24b4b0c6b3de27568a98b41f5e59eafd8c060dee848992c4
vendor/mldsa-native/dev/x86_64/src/poly_caddq_avx2.c	1609	07a5d8faa06b24399aed68397059063a4afbc8fe8ebedd903f9aa
vendor/mldsa-native/dev/x86_64/src/poly_chknorm_avx2.c	1271	366ff5b9bac1e0d0cf3aedecbc5f3de0ee9677c9fd940091b7ae0
vendor/mldsa-native/dev/x86_64/src/poly_decompose_32_avx2.c	5971	a0cf2f3a88c685d1d8f78799baa6aa8878e54a17144b4fe3c79b8
vendor/mldsa-native/dev/x86_64/src/poly_decompose_88_avx2.c	5871	2a389287fc5ac137852aad1373f65cc11da9f178063296876f70b
vendor/mldsa-native/dev/x86_64/src/poly_use_hint_32_avx2.c	3690	87dccba1100f78804576cd41826e13ccc4598e0d36f6fc031d67a
vendor/mldsa-native/dev/x86_64/src/poly_use_hint_88_avx2.c	3673	00d6bd43e97a75c44da4adebd314aadbe77dc31c2889489aadb33
vendor/mldsa-native/dev/x86_64/src/polyz_unpack_17_avx2.c	3834	16d0f450551f80edecd5c26fa6025ff54532589a0ca4360cff09f
vendor/mldsa-native/dev/x86_64/src/polyz_unpack_19_avx2.c	4005	2c1a08402202bad2183b346fa262a88de442f0164bb25d1c1447e
vendor/mldsa-native/dev/x86_64/src/rej_uniform_avx2.c	6353	f99f891e5c98bcef846d38925d053262851536b589dd7f0d3c287
vendor/mldsa-native/dev/x86_64/src/rej_uniform_eta2_avx2.c	4574	7056af7757693807ac3efe1653dad224049ce5985974a4e14518d
vendor/mldsa-native/dev/x86_64/src/rej_uniform_eta4_avx2.c	3936	42d382f31be9a4c934b6dd1142deb8574fdd6f4f8f41d03ea6bea
vendor/mldsa-native/dev/x86_64/src/rej_uniform_table.c	7960	d6caed3dd9d9cf6b1391cfc9537d31a84384b6bcf380cd0391a43

vendor/mldsa-native/integration/liboqs/config_aarch64.h	13321	d603eafb0fb7e5069e3013f01c9bdbb39b1ac70ad12b582cbaf22
vendor/mldsa-native/integration/liboqs/config_c.h	13408	956af96ce26cb46c1010235f204439e9de2f92a297f8ea7fe8fb9
vendor/mldsa-native/integration/liboqs/config_x86_64.h	13314	9807424b0baeebbe19666a217b491ba45adb34dd69b6c30c6b807
vendor/mldsa-native/integration/liboqs/fips202_glue.h	923	b7174a806dec586fa6128b0f85d7bf611161502f788499de1779f
vendor/mldsa-native/integration/liboqs/fips202x4_glue.h	987	b988cb80028187028a51f2a90ec427b870ae02d772f6a450d966b
vendor/mldsa-native/mldsa/mldsa_native.c	22252	cea0f7cd3c72998bf28762b906ed0b103cbcd911f59f0b080b763
vendor/mldsa-native/mldsa/mldsa_native.h	37676	01ae99fa0e65a17f9481573af8c2dbea3bc6cb63dc0594610805a
vendor/mldsa-native/mldsa/mldsa_native_config.h	31211	7464f5fda2bb3b0baf368507c0344acb0b343a68864da5880aca9
vendor/mldsa-native/mldsa/src/cbmc.h	6227	328ed11c0652bc5bb3b58d1650d03190a83985e11f0817e84f483
vendor/mldsa-native/mldsa/src/common.h	12294	5d4a056613b6ea8d8bec40af7838df48ffd079047a44747268567
vendor/mldsa-native/mldsa/src/ct.c	660	5d5641f89ee92f8d33f4489b7746ca20d4226cf6ecfcc9895bf64
vendor/mldsa-native/mldsa/src/ct.h	12552	95ef3775caabfd1e222b50ebea5c963754d6c270f8d5980065d85
vendor/mldsa-native/mldsa/src/debug.c	1765	14c8356b7cd6ad0b6de8e2915468aa5271bcea2c2d959cfe22180
vendor/mldsa-native/mldsa/src/debug.h	5172	dbea92aa5b1a8b9afa09eb93bbf2532f5bb8656b7dc211a331e31
vendor/mldsa-native/mldsa/src/fips202/fips202.c	8925	be2e087faac448a4a07f6680f569c890f91b365f00fbbc25caedd
vendor/mldsa-native/mldsa/src/fips202/fips202.h	8890	29550569a1439261e6f587cadf18e956308c87e3c64f8f8d01119
vendor/mldsa-native/mldsa/src/fips202/fips202x4.c	6470	0d97a2a25883f20a321c451ac77e3b8fa2683051e478adfc97900
vendor/mldsa-native/mldsa/src/fips202/fips202x4.h	4429	5dc714bd8d106b8ce850e6ff2b5da0ad1eca92afc78aefa6d2e62
vendor/mldsa-native/mldsa/src/fips202/keccakf1600.c	12757	cd00ab2e66c80423ee597ca18aabe83f1c4c90cc2f44defd44c3a
vendor/mldsa-native/mldsa/src/fips202/keccakf1600.h	4522	9bd102a113dfe2799ff47b3755e14d69d5cc963d14e71a54a8552
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/auto.h	1997	df06198063437d5a83e5c99ae68cbdd221f402175ef4e7055b931
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/src/fips202_native_aarch64.h	2487	531fe8d1ec62897118f8df08ec83ff419f5772ef040d8c6f147d4
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/src/keccakf1600_round_constants.c	1996	c9351871c008c57156ea5b5192e39ffbef7dc617bd944c640ef8b
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/x1_scalar.h	766	410b7e592502bd123cbb6564da582a205ed46fb15bf10115e3e27
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/x1_v84a.h	965	38752a44470fc1a70ec29510f90c427fa1e9a47c6aa855211e667
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/x2_v84a.h	1056	8976244b9368e6e66c42f7f9ed62ae0fe42edf5766694930cdc79
vendor/mldsa-native/mldsa/src/fips202/native/aarch64/x4_v8a_scalar.h	844	9cbdf46a0e9264a60bf7b024482a15fabf17a52de5533cd39d359

vendor/mldsa-native/mldsa/src/fips202/native/aarch64/x4_v8a_v84a_scalar.h	1041	987cc6617778df6152474d0a7529a6395b16ee67a9386ea39767c
vendor/mldsa-native/mldsa/src/fips202/native/api.h	2812	8b3fc53dcfdd088b73519f274b6713aa370c00af98addc831eba7
vendor/mldsa-native/mldsa/src/fips202/native/armv81m/mve.h	822	5ff69b17de89d246d6a13eba733a48e81b793d662007fac8aaaf6
vendor/mldsa-native/mldsa/src/fips202/native/armv81m/src/fips202_native_armv81m.h	803	e629cabe779960e496858cd63e91ea6e599e0e0af1432927352f3
vendor/mldsa-native/mldsa/src/fips202/native/armv81m/src/keccak_f1600_x4_mve.c	4687	03dd17852b4827d662ff622de4070b392ca5c590ef36de0cad9d6
vendor/mldsa-native/mldsa/src/fips202/native/armv81m/src/keccakf1600_round_constants.c	1729	9b8bf66e1bbf69d8ada688d0ebb029a4b57d1053ce3b0f6767240
vendor/mldsa-native/mldsa/src/fips202/native/auto.h	699	dc3607081d516a8d1310465a48984cb223ee4d5d1b78966304714
vendor/mldsa-native/mldsa/src/fips202/native/x86_64/src/KeccakP_1600_times4_SIMD256.c	28194	3209f56295093fdb9749ee0639e3bc50c12cfa495ea537d557b2d
vendor/mldsa-native/mldsa/src/fips202/native/x86_64/src/KeccakP_1600_times4_SIMD256.h	547	54e7fe6f1bfbcf042888c82758ea8395bdd8b020fcba49109da7f
vendor/mldsa-native/mldsa/src/fips202/native/x86_64/xkcp.h	777	8c56fa9a1fe025d35627cf6f48efad820ab4803e3600a099bd203
vendor/mldsa-native/mldsa/src/native/aarch64/meta.h	9372	86458b409219c5b521acd1c8e780a7dd5b680ab1607f548861cbf
vendor/mldsa-native/mldsa/src/native/aarch64/src/aarch64_zetas.c	14368	87e22e4dd92e86f655870785fc8a6f40be187b543c0466920092f
vendor/mldsa-native/mldsa/src/native/aarch64/src/arith_native_aarch64.h	6497	406b41271328f5f5488b25a2dd061787c2d73bc4e1cc1e0b13ce7
vendor/mldsa-native/mldsa/src/native/aarch64/src/polyz_unpack_table.c	1417	0961cb66fd4951a3b01a243cf6886949d497b7ed19fefc7264d45
vendor/mldsa-native/mldsa/src/native/aarch64/src/rej_uniform_eta_table.c	25597	9bd3425816dcc0c21115f6955cc8830c4cdc3c7bebb1e1d6ed76e
vendor/mldsa-native/mldsa/src/native/aarch64/src/rej_uniform_table.c	2295	3308702a2208ea746e77c02792fa47ccfd0ea7ed7fc439e7cbcfd
vendor/mldsa-native/mldsa/src/native/api.h	31647	bef56a3797acf146c933f5f8bb759491ac1d9247992170cbe65a8
vendor/mldsa-native/mldsa/src/native/meta.h	421	bb197e13c58a1a71d7597cb5224b5e949ebc800463fe89ff28e06
vendor/mldsa-native/mldsa/src/native/x86_64/meta.h	10542	5a629bba590fc8b7f8ade17e07a7e28ad93cf2e7007d8bf91fcbf
vendor/mldsa-native/mldsa/src/native/x86_64/src/arith_native_x86_64.h	5074	e52ef79ddefd09e1bf941a2794435393e8e2cd8c9e1916504342b
vendor/mldsa-native/mldsa/src/native/x86_64/src/consts.c	9085	25ab5b8f9def84990da5cd747f2ff2298c684c6c6fc2441ed79c4
vendor/mldsa-native/mldsa/src/native/x86_64/src/consts.h	806	9f19c24b4b0c6b3de27568a98b41f5e59eafd8c060dee848992c4
vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_caddq_avx2.c	1609	07a5d8faa06b24399aed68397059063a4afbc8fe8ebedd903f9aa

vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_chknorm_avx2.c	1271	366ff5b9bac1e0d0cf3aedecbc5f3de0ee9677c9fd940091b7ae0
vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_decompose_32_avx2.c	5971	a0cf2f3a88c685d1d8f78799baa6aa8878e54a17144b4fe3c79b8
vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_decompose_88_avx2.c	5871	2a389287fc5ac137852aad1373f65cc11da9f178063296876f70b
vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_use_hint_32_avx2.c	3690	87dccba1100f78804576cd41826e13ccc4598e0d36f6fc031d67a
vendor/mldsa-native/mldsa/src/native/x86_64/src/poly_use_hint_88_avx2.c	3673	00d6bd43e97a75c44da4adebd314aadbe77dc31c2889489aadb33
vendor/mldsa-native/mldsa/src/native/x86_64/src/polyz_unpack_17_avx2.c	3834	16d0f450551f80edecd5c26fa6025ff54532589a0ca4360cff09f
vendor/mldsa-native/mldsa/src/native/x86_64/src/polyz_unpack_19_avx2.c	4005	2c1a08402202bad2183b346fa262a88de442f0164bb25d1c1447e
vendor/mldsa-native/mldsa/src/native/x86_64/src/rej_uniform_avx2.c	6353	f99f891e5c98bcef846d38925d053262851536b589dd7f0d3c287
vendor/mldsa-native/mldsa/src/native/x86_64/src/rej_uniform_eta2_avx2.c	4574	7056af7757693807ac3efe1653dad224049ce5985974a4e14518d
vendor/mldsa-native/mldsa/src/native/x86_64/src/rej_uniform_eta4_avx2.c	3936	42d382f31be9a4c934b6dd1142deb8574fdd6f4f8f41d03ea6bea
vendor/mldsa-native/mldsa/src/native/x86_64/src/rej_uniform_table.c	7960	d6caed3dd9d9cf6b1391cfc9537d31a84384b6bcf380cd0391a43
vendor/mldsa-native/mldsa/src/packing.c	9210	a7fad3acf40f0b1f18813a5cad6066e3b84956bbcdc8fc7235e2c
vendor/mldsa-native/mldsa/src/packing.h	9708	9852b2289f3b4172473423f6c15c25b9ad6dd57be8b987ad31f55
vendor/mldsa-native/mldsa/src/params.h	2176	9752e4fb21ed390fc94cd9d05691e5c1b76939a69b7c7ed72f8e1
vendor/mldsa-native/mldsa/src/poly.c	35409	069577780908dc2d06a21556365712e8408340dfc0cdab39b68ce
vendor/mldsa-native/mldsa/src/poly.h	16291	3b714379ae31e640dea735a24b3442e3e76096874e6739b67f133
vendor/mldsa-native/mldsa/src/poly_kl.c	34192	5a4330471e2811364d67e2826ad94fa9b4ae69013f7357f21a7bf
vendor/mldsa-native/mldsa/src/poly_kl.h	15723	93bd01ec8d5f2e7f8e8c2cdff3e2a006276b198f255c7f58ebe5d
vendor/mldsa-native/mldsa/src/polyvec.c	29352	462482045e569d2a96df2bf829daba1900b816995831f6d9b3d16
vendor/mldsa-native/mldsa/src/polyvec.h	31718	547e679719c0e9adf02b9dd855946e700f368db556893b3fb4524
vendor/mldsa-native/mldsa/src/randombytes.h	730	b8ccb9a34818d8e1cd0cfea9b24829ab9949cb88c20ed0648459c
vendor/mldsa-native/mldsa/src/reduce.h	4421	6bdce26e08c8d8895f242d6be2cd7145a5db227297a43ce161db5
vendor/mldsa-native/mldsa/src/rounding.h	8980	e32638772ee4564c683685b7f9acc84e5e0763e6ca3eb61abb7ed
vendor/mldsa-native/mldsa/src/sign.c	53437	b9ba66f68ba63d1462607cbc23aa7f66203084769dc2ab0ecfae3
vendor/mldsa-native/mldsa/src/sign.h	37165	6c3dc7a4e151257cff1dd30ed58c47c48886fa42fab69c178180c
vendor/mldsa-native/mldsa/src/symmetric.h	2635	4ed2df211b960a747e29aec04055100470def1ae8c481feb203c2

vendor/mldsa-native/mldsa/src/sys.h	8233	e06262e46cccab6479f296bea6ec25f7efe4fc0901dcca6fcc6c9
vendor/mldsa-native/proofs/cbmc/H/H_harness.c	527	e749847d3f8c767b90624626b37b0c4af774eade89a9bf964c757
vendor/mldsa-native/proofs/cbmc/attempt_signature_generation/attempt_signature_generation_harness.c	831	ae6995401b38cca523bba39b83d5680a0b992dbad065337c9df8c
vendor/mldsa-native/proofs/cbmc/caddq/caddq_harness.c	194	f9d8824447cd28249f74839921d89964a3f2ec6e3123cc23fd74f
vendor/mldsa-native/proofs/cbmc/check_pct/check_pct_harness.c	370	689efda6dda3e3df8be93861b8603f636edb345170008653fc8fe
vendor/mldsa-native/proofs/cbmc/compute_pack_z/compute_pack_z_harness.c	471	3c08fbd60f4b4f5ab3c4188dd541d06322bb860b029c7454af62a
vendor/mldsa-native/proofs/cbmc/compute_t0_t1_tr_from_sk_components/compute_t0_t1_tr_from_sk_components_harness.c	592	a544301effa169155e36e7f65ce05d959376dcde387b97d8f9422
vendor/mldsa-native/proofs/cbmc/ct_abs_i32/ct_abs_i32_harness.c	178	e66eae9fc96dc3bcf4360c8be26126e06a0a4c59adaf5e1a4992e
vendor/mldsa-native/proofs/cbmc/ct_cmask_neg_i32/ct_cmask_neg_i32_harness.c	184	ed9551addb563305fa9cbec757a7d77ccf74b14948313a32e29b7
vendor/mldsa-native/proofs/cbmc/ct_cmask_nonzero_u32/ct_cmask_nonzero_u32_harness.c	313	331d478700ba98d10c129f7a25e9985c6cfed6d9e72872f33e2ec
vendor/mldsa-native/proofs/cbmc/ct_cmask_nonzero_u8/ct_cmask_nonzero_u8_harness.c	311	6bc2b61cededf0e83ba25b14c673e8431ada042b6a794ecd03288
vendor/mldsa-native/proofs/cbmc/ct_get_optblocker_i64/ct_get_optblocker_i64_harness.c	173	5668162b8a1efca7c18f3ca85863c69934b819702c3c94d6f6200
vendor/mldsa-native/proofs/cbmc/ct_get_optblocker_u32/ct_get_optblocker_u32_harness.c	173	d049b9acb822c220941d4ff114b1f7e8f2b558341e45ab6e0db0f
vendor/mldsa-native/proofs/cbmc/ct_get_optblocker_u8/ct_get_optblocker_u8_harness.c	250	b7808efafa87e3ea2277f24b3acf8cc1a19a94b7547add727c16f
vendor/mldsa-native/proofs/cbmc/ct_memcmp/ct_memcmp_harness.c	293	8d9228d349de84dda94943610525af23819275c20427c7d6c366e
vendor/mldsa-native/proofs/cbmc/ct_sel_int32/ct_sel_int32_harness.c	209	3b5b44ba8264a5521744b3db9a2e3f4fd052ae277f91daffe67c1
vendor/mldsa-native/proofs/cbmc/decompose/decompose_harness.c	212	56a8d8087f70bf7df54ba5d17f6fe3cfc284fd3d53042522fcec2
vendor/mldsa-native/proofs/cbmc/dummy_backend.h	1007	bd8e59bf29c6de7f0cad6063473a52e447d18af2d4b0a428ae2a0
vendor/mldsa-native/proofs/cbmc/dummy_backend_fips202_x1.h	317	96b98dc14759349d4df9b22f64e14c3d0b115effb7f0bfd4c16b0

vendor/mldsa-native/proofs/cbmc/dummy_backend_fips202_x4.h	317	cde24902521e1f5b71bb18d8b9180920760089b098916faa82aaa
vendor/mldsa-native/proofs/cbmc/fqmul/fqmul_harness.c	229	959b03e64a88f12451edb6c20566cdfb39d67143e998a93bb0642
vendor/mldsa-native/proofs/cbmc/fqscale/fqscale_harness.c	216	75172268b10d4eb6438afab0da048e1d85d05aaeb60271a57e156
vendor/mldsa-native/proofs/cbmc/intt_native_x86_64/intt_native_x86_64_harness.c	281	b5c8189ba6f1fac79794453e916ccf64e9a11e5af527b5bcd8efe
vendor/mldsa-native/proofs/cbmc/invntt_layer/invntt_layer_harness.c	290	b6decf5f9c7b946d36a3584dceb636f61aa32f8eba47f0b5bc2b9
vendor/mldsa-native/proofs/cbmc/keccak_absorb/keccak_absorb_harness.c	494	c8b604a618b2341f278ac085aa0387d6735c7efa63ffb2f7caf54
vendor/mldsa-native/proofs/cbmc/keccak_absorb_once_x4/keccak_absorb_once_x4_harness.c	703	0ab6a116e2e7b4d0e17d33bef4e9e691ea923e3b25703c02f8ec2
vendor/mldsa-native/proofs/cbmc/keccak_finalize/keccak_finalize_harness.c	375	4147bf5ceb790eb7403de0d2a0d2d045107d6a5b550693042f251
vendor/mldsa-native/proofs/cbmc/keccak_init/keccak_init_harness.c	247	d934019cc0e71d9eeaa62fc1a5e014f5fd5afe7b3d4435f842274
vendor/mldsa-native/proofs/cbmc/keccak_squeeze/keccak_squeeze_harness.c	471	cd602700f126ea46245fb11b92789cf213dfac73a7d4fd199d9c2
vendor/mldsa-native/proofs/cbmc/keccak_squeezeblocks_x4/keccak_squeezeblocks_x4_harness.c	669	b57affd2cf79a582ecd3aa72be6b63819908a3f6c88b8294aa3be
vendor/mldsa-native/proofs/cbmc/keccakf1600_extract_bytes/keccakf1600_extract_bytes_harness.c	325	1e1a3d064aa57342f541931970cd49198e801b059c6451f427b18
vendor/mldsa-native/proofs/cbmc/keccakf1600_extract_bytes_BE/keccakf1600_extract_bytes_be_harness.c	325	1e1a3d064aa57342f541931970cd49198e801b059c6451f427b18
vendor/mldsa-native/proofs/cbmc/keccakf1600_permute/keccakf1600_permute_harness.c	198	c844e9144ab782ec5604ade65ea925af1f2951cb7e80ff42e0a4c
vendor/mldsa-native/proofs/cbmc/keccakf1600_permute_native/keccakf1600_permute_native_harness.c	198	c844e9144ab782ec5604ade65ea925af1f2951cb7e80ff42e0a4c
vendor/mldsa-native/proofs/cbmc/keccakf1600_xor_bytes/keccakf1600_xor_bytes_harness.c	398	96d61e662fc375f9586f9d2d50092365630fb839ae464ac862e9d
vendor/mldsa-native/proofs/cbmc/keccakf1600_xor_bytes_BE/keccakf1600_xor_bytes_be_harness.c	398	96d61e662fc375f9586f9d2d50092365630fb839ae464ac862e9d

vendor/mldsa-native/proofs/cbmc/keccakf1600x4_extract_bytes/keccakf1600x4_extract_bytes_harness.c	479	cde3591d13269dbe5fe0071b1f9a321cd8c3f91fdf5070be6024f
vendor/mldsa-native/proofs/cbmc/keccakf1600x4_permute/keccakf1600x4_permute_harness.c	301	32d13750d6249c35b8b49f1b24c96ff15456c3ac39e812d04a34b
vendor/mldsa-native/proofs/cbmc/keccakf1600x4_permute_native/keccakf1600x4_permute_native_harness.c	301	32d13750d6249c35b8b49f1b24c96ff15456c3ac39e812d04a34b
vendor/mldsa-native/proofs/cbmc/keccakf1600x4_xor_bytes/keccakf1600x4_xor_bytes_harness.c	477	19d4fa4254cf02f0415ad022f53fdb118b2e7c1f5af3cbe06301c
vendor/mldsa-native/proofs/cbmc/lib/__init__.py	52	34743113a2b63e6ac66a6019ce7182327c3435bbfdbe6486b17bf
vendor/mldsa-native/proofs/cbmc/lib/print_tool_versions.py	1875	b0ab0604a9d785964bcf05c344564361567567fcd076c711a475f
vendor/mldsa-native/proofs/cbmc/lib/summarize.py	7122	d19d04bae53d0b7cf7aaa174592015a15e7f15efc9a0866ae2438
vendor/mldsa-native/proofs/cbmc/make_hint/make_hint_harness.c	211	e0e3c615e0e9ac954b545504c3428662d403adc5098e703068590
vendor/mldsa-native/proofs/cbmc/mldsa_native_config_cbmc.h	31575	27362b86f5896bdd16fecc5f31070ab5adb3fdd477e67580ee454
vendor/mldsa-native/proofs/cbmc/montgomery_reduce/montgomery_reduce_harness.c	206	d3ec1f2d7a3c236a4d8b15171944fa1887cdd8930660d1bbccfd7
vendor/mldsa-native/proofs/cbmc/ntt_butterfly_block/ntt_butterfly_block_harness.c	424	8f0fa4eeab6935b7314d7652856bb8b1a374f124da1b28272eb35
vendor/mldsa-native/proofs/cbmc/ntt_layer/ntt_layer_harness.c	284	42b103c069ef26d6453ffb8acb4ed693dd71fcac4e3f14db4723e
vendor/mldsa-native/proofs/cbmc/ntt_native_aarch64/ntt_native_aarch64_harness.c	279	831338b93557f55f6e8f8bc5d68da5f30f0ca6f2899b20d0c3d63
vendor/mldsa-native/proofs/cbmc/ntt_native_x86_64/ntt_native_x86_64_harness.c	279	831338b93557f55f6e8f8bc5d68da5f30f0ca6f2899b20d0c3d63
vendor/mldsa-native/proofs/cbmc/pack_pk/pack_pk_harness.c	211	5f5464fe13fd1a13c39a9fea93bda4fe6e33cc01c069838b196f3
vendor/mldsa-native/proofs/cbmc/pack_sig_c_h/pack_sig_c_h_harness.c	239	d90dd7ebac29c2560ea9c2c07e89ceaa2814f9c89516e7d3a3706
vendor/mldsa-native/proofs/cbmc/pack_sig_z/pack_sig_z_harness.c	222	8543f988096f9dd4fc3510cb287359e644257593c7dfdccac600e
vendor/mldsa-native/proofs/cbmc/pack_sk/pack_sk_harness.c	260	419f9e42d6191553ec99b6345a7bfb8ab729d5645b86cb7280adf
vendor/mldsa-native/proofs/cbmc/poly_add/poly_add_harness.c	187	692027b67f1a5d054cc7078c475d268b05a827baae2fa7afb3382
vendor/mldsa-native/proofs/cbmc/poly_caddq/poly_caddq_harness.c	183	6b0389b3d65f668c64e318bf1682fb405cab925d407d1ee71e34f

vendor/mldsa-native/proofs/cbmc/poly_caddq_c/poly_caddq_c_harness.c	263	259862d68ad758d412dbacb668ece357b1d5e19bea7632f2b77e8
vendor/mldsa-native/proofs/cbmc/poly_caddq_native/poly_caddq_native_harness.c	183	6b0389b3d65f668c64e318bf1682fb405cab925d407d1ee71e34f
vendor/mldsa-native/proofs/cbmc/poly_caddq_native_aarch64/poly_caddq_native_aarch64_harness.c	289	5f2f7e02d0661fc34bbab6c41d7fff957937eb83a659df9249425
vendor/mldsa-native/proofs/cbmc/poly_challenge/poly_challenge_harness.c	213	18e7f7451aaac811149c8c305c1c7d96434fd5e3a0abdd772bc37
vendor/mldsa-native/proofs/cbmc/poly_chknorm/poly_chknorm_harness.c	222	edef1c4ef04e1ba7c42bc8a7e1dcc8ca761ceb216017c7757ffcd
vendor/mldsa-native/proofs/cbmc/poly_chknorm_c/poly_chknorm_c_harness.c	318	bf1b411bfbadd8bec4775371e0260617af10852801faa4eb504a7
vendor/mldsa-native/proofs/cbmc/poly_chknorm_native/poly_chknorm_native_harness.c	222	edef1c4ef04e1ba7c42bc8a7e1dcc8ca761ceb216017c7757ffcd
vendor/mldsa-native/proofs/cbmc/poly_chknorm_native_aarch64/poly_chknorm_native_aarch64_harness.c	332	b57e5b09592aa320aa81221a2e74becb22bd28057b31147b0d6d3
vendor/mldsa-native/proofs/cbmc/poly_decompose/poly_decompose_harness.c	201	7b4e7156247c75e7395be5bad8220fa8654d10543f257cd1d865f
vendor/mldsa-native/proofs/cbmc/poly_decompose_c/poly_decompose_c_harness.c	368	65ec91efab88016745e454c3a6739cf052ea484d0587e6491b5bf
vendor/mldsa-native/proofs/cbmc/poly_decompose_native/poly_decompose_native_harness.c	201	7b4e7156247c75e7395be5bad8220fa8654d10543f257cd1d865f
vendor/mldsa-native/proofs/cbmc/poly_invntt_tomont/poly_invntt_tomont_harness.c	230	907160eb7f5dd06ca0ee79ac442783b5eed7ffb14b5db100241ba
vendor/mldsa-native/proofs/cbmc/poly_invntt_tomont_c/poly_invntt_tomont_c_harness.c	319	36c15a1fd269700c7a3ccb94421a098d80fde05db2ea30c068f08
vendor/mldsa-native/proofs/cbmc/poly_invntt_tomont_native/poly_invntt_tomont_native_harness.c	190	289370c115290baf53c6d099e9a2d484ae64b64b2bee805fc995d
vendor/mldsa-native/proofs/cbmc/poly_make_hint/poly_make_hint_harness.c	225	42bd17c27d93686c3aab69275becd2e1a9493e6ded3e3a71a7052
vendor/mldsa-native/proofs/cbmc/poly_ntt/poly_ntt_harness.c	180	30de42fc7799dbc347dc1a0398182f378699b0f798a3b102cb05d
vendor/mldsa-native/proofs/cbmc/poly_ntt_c/poly_ntt_c_harness.c	258	3a4b83c31c13e0244228f831d55133138a77aef24b47af5e69307
vendor/mldsa-native/proofs/cbmc/poly_ntt_native/poly_ntt_native_harness.c	180	30de42fc7799dbc347dc1a0398182f378699b0f798a3b102cb05d
vendor/mldsa-native/proofs/cbmc/poly_pointwise_montgomery/poly_pointwise_montgomery_harness.c	211	9e9a87f3dece9742c166a0907d900ae5c889c919996197c31b664

vendor/mldsa-native/proofs/cbmc/poly_pointwise_montgomery_c/poly_pointwise_montgomery_c_harness.c	333	d53b3d205f1712e6a9643a0408952aedc3f5cd92866fd54a8a47e
vendor/mldsa-native/proofs/cbmc/poly_pointwise_montgomery_native/poly_pointwise_montgomery_native_harness.c	211	9e9a87f3dece9742c166a0907d900ae5c889c919996197c31b664
vendor/mldsa-native/proofs/cbmc/poly_power2round/poly_power2round_harness.c	207	881bedd316a96fb7d3c4cb8b70bb2aa3f0377f18617cb5dcdae73
vendor/mldsa-native/proofs/cbmc/poly_reduce/poly_reduce_harness.c	183	588953c052099600f23be51a09e1c812222a525cf8594e91c8c63
vendor/mldsa-native/proofs/cbmc/poly_shiftl/poly_shiftl_harness.c	183	0c1f24d94e6c978cca90a0d34a029f50eefe4b282e1ad01b3efdb
vendor/mldsa-native/proofs/cbmc/poly_sub/poly_sub_harness.c	187	334af03d7558ea47c76d1c939b66f86216c6191440e7f02aeb7ed
vendor/mldsa-native/proofs/cbmc/poly_uniform/poly_uniform_harness.c	243	6d739549c4e484a3145e4802a5a6b70b6b58c6f9bda93c95a59cb
vendor/mldsa-native/proofs/cbmc/poly_uniform_4x/poly_uniform_4x_harness.c	310	f2cddb38b31e53d17ac20dda7e81a386e031009c747818d7cc96e
vendor/mldsa-native/proofs/cbmc/poly_uniform_eta/poly_uniform_eta_harness.c	241	6f2554154a41c68d5057bb9903bf4e8ff74254073c3199786499d
vendor/mldsa-native/proofs/cbmc/poly_uniform_eta_4x/poly_uniform_eta_4x_harness.c	295	9b8a74c5e7496bd197507cb19780c79232f26bd5103cd3e327050
vendor/mldsa-native/proofs/cbmc/poly_uniform_gamma1/poly_uniform_gamma1_harness.c	388	232cbaca6d3c1f0d3f1b6d3901a1fd8ea18295125f3b2ef2833f4
vendor/mldsa-native/proofs/cbmc/poly_uniform_gamma1_4x/poly_uniform_gamma1_4x_harness.c	299	58b0068d870dddb94ba045e5f7c85365ff95f31dd1df664f07b45
vendor/mldsa-native/proofs/cbmc/poly_use_hint/poly_use_hint_harness.c	203	7861d11cf75f2b52050226a1310847612ef0bfb3266237c5dd578
vendor/mldsa-native/proofs/cbmc/poly_use_hint_c/poly_use_hint_c_harness.c	379	e40f57748b95d4d4f1efdae4cad864f34e75cee1fcb11050266bd
vendor/mldsa-native/proofs/cbmc/poly_use_hint_native/poly_use_hint_native_harness.c	203	7861d11cf75f2b52050226a1310847612ef0bfb3266237c5dd578
vendor/mldsa-native/proofs/cbmc/polyeta_pack/polyeta_pack_harness.c	204	9f2346fbb8acdb498cab30f1b66f766ca0879360a874bf4ade603
vendor/mldsa-native/proofs/cbmc/polyeta_unpack/polyeta_unpack_harness.c	206	e797b287b8e43be91d8b327baf8073def1f6e6ecb46a060dd2d59
vendor/mldsa-native/proofs/cbmc/polymat_permute_bitrev_to_custom/polymat_permute_bitrev_to_custom_harness.c	276	5246e2088be4e55ec041752d3cd4e12e97bcd8f7e6cd26214cd84
vendor/mldsa-native/proofs/cbmc/polyt0_pack/polyt0_pack_harness.c	200	0cc9c4c79e88d13dab0d7050fdc4d95fdc02b0f442ae5349177d5

vendor/mldsa-native/proofs/cbmc/polyt0_unpack/polyt0_unpack_harness.c	202	51b72eff6cedcbc85af6e65f30e71cdc83b267888e1532c619e12
vendor/mldsa-native/proofs/cbmc/polyt1_pack/polyt1_pack_harness.c	200	c5ff9ac76197959cf122094b52b10b7abd4b373a46791cf8a48b6
vendor/mldsa-native/proofs/cbmc/polyt1_unpack/polyt1_unpack_harness.c	202	55eaf2a62d5e2d8c74f654ded9473c192b76c386974217f626315
vendor/mldsa-native/proofs/cbmc/polyvec_matrix_expand/polyvec_matrix_expand_harness.c	225	40c33a9fe78ca444fbf20cf1d6dce8b7e73e2bfa53e9f76611aaf
vendor/mldsa-native/proofs/cbmc/polyvec_matrix_expand_serial/polyvec_matrix_expand_harness.c	225	40c33a9fe78ca444fbf20cf1d6dce8b7e73e2bfa53e9f76611aaf
vendor/mldsa-native/proofs/cbmc/polyvec_matrix_pointwise_montgomery/polyvec_matrix_pointwise_montgomery_harness.c	257	6d863d6f846cb42ee9af68287dab732b0d6bc484f62d3a7026267
vendor/mldsa-native/proofs/cbmc/polyveck_add/polyveck_add_harness.c	198	02c2d04095b6417fd055b699122179cc14c225d39b1627da0e2e6
vendor/mldsa-native/proofs/cbmc/polyveck_caddq/polyveck_caddq_harness.c	193	17e5fa581ada0f479e233fa24c7e00e5bd40f766de9720a88e9bd
vendor/mldsa-native/proofs/cbmc/polyveck_chknorm/polyveck_chknorm_harness.c	229	4af1d58a2a0683ba21e47a0576aa41d36ca001d81bbb5ef49746d
vendor/mldsa-native/proofs/cbmc/polyveck_decompose/polyveck_decompose_harness.c	208	44fc31ba646a62f36088f9bda37d2ba9435d70aa128f8671eecc3
vendor/mldsa-native/proofs/cbmc/polyveck_invntt_tomont/polyveck_invntt_tomont_harness.c	241	7b13da771afe678e3edfdc22876f02871d256774c4229c6279576
vendor/mldsa-native/proofs/cbmc/polyveck_make_hint/polyveck_make_hint_harness.c	233	db53736caf25b8e189b2c727a643843708f8e4d9b990e46595f7d
vendor/mldsa-native/proofs/cbmc/polyveck_ntt/polyveck_ntt_harness.c	191	e40816a6c1925779ef2300db035f28cc46cd2fe497b05d1f446ab
vendor/mldsa-native/proofs/cbmc/polyveck_pack_eta/polyveck_pack_eta_harness.c	213	766c08d262e5b5d87e530ef780dc4b41cb8b5f34f439836911503
vendor/mldsa-native/proofs/cbmc/polyveck_pack_t0/polyveck_pack_t0_harness.c	212	f9d2d448c4c5e5fe00270dfc2b7a9466cc04463e92da2062e3f13
vendor/mldsa-native/proofs/cbmc/polyveck_pack_w1/polyveck_pack_w1_harness.c	212	998802abc5584ee719867e2723e4915aea25415e9e33878a8aa19
vendor/mldsa-native/proofs/cbmc/polyveck_pointwise_poly_montgomery/polyveck_pointwise_poly_montgomery_harness.c	238	eba56283557e023eb4306656c1fc8830d9d1292bfce37ae87de27
vendor/mldsa-native/proofs/cbmc/polyveck_power2round/polyveck_power2round_harness.c	213	0a0fdfdbb62786d3840cf5344395c3ebe8132424f2f63b8b512d2

vendor/mldsa-native/proofs/cbmc/polyveck_reduce/polyveck_reduce_harness.c	194	6c46aa82cf2965d7d72b62054cf16ff2d62a823e519ba584e550d
vendor/mldsa-native/proofs/cbmc/polyveck_shiftl/polyveck_shiftl_harness.c	194	96b4c489dd75a37d64ecf41cd8e8a444dce076488dd72052ec83d
vendor/mldsa-native/proofs/cbmc/polyveck_sub/polyveck_sub_harness.c	198	f1d0467f63cf2c521da6565547fffbcff2c7a7ca1bf362c26426c
vendor/mldsa-native/proofs/cbmc/polyveck_unpack_eta/polyveck_unpack_eta_harness.c	215	d1bba1b5fca09379db8d0bc909707bc5459a6d0dd27ec7e7460b0
vendor/mldsa-native/proofs/cbmc/polyveck_unpack_t0/polyveck_unpack_t0_harness.c	214	7921912bb4dd500324da3f04d256c7a0d21fe93634095ac63000d
vendor/mldsa-native/proofs/cbmc/polyveck_use_hint/polyveck_use_hint_harness.c	210	f76c9f42a029dfac1342fffc976f18fc98fb7c9e76e12c98c6ae8
vendor/mldsa-native/proofs/cbmc/polyvecl_chknorm/polyvecl_chknorm_harness.c	220	12779c60b393ed168aa82ed17370832329580f4a459c287e310ea
vendor/mldsa-native/proofs/cbmc/polyvecl_ntt/polyvecl_ntt_harness.c	191	adf108ee1420e3908e1e670636caba6678df1ab5f5251edab814f
vendor/mldsa-native/proofs/cbmc/polyvecl_pack_eta/polyvecl_pack_eta_harness.c	213	e3e34ac585455e38de3fa8b4223134a0e6fab1596e3698ecb83d4
vendor/mldsa-native/proofs/cbmc/polyvecl_permute_bitrev_to_custom/polyvecl_permute_bitrev_to_custom_harness.c	274	362188cffed11abeacb7f6eafa50fff67ae01e7813c8d9d5deb76
vendor/mldsa-native/proofs/cbmc/polyvecl_permute_bitrev_to_custom_native/polyvecl_permute_bitrev_to_custom_native_harness.c	274	362188cffed11abeacb7f6eafa50fff67ae01e7813c8d9d5deb76
vendor/mldsa-native/proofs/cbmc/polyvecl_pointwise_acc_montgomery/polyvecl_pointwise_acc_montgomery_harness.c	237	25f0386652609a7b56fc7ea3ccb52fd753de0bc0cc9d95a387554
vendor/mldsa-native/proofs/cbmc/polyvecl_pointwise_acc_montgomery_c/polyvecl_pointwise_acc_montgomery_c_harness.c	531	139d2a314a5f2c508483dfe9ad3fa7d02b64bc238840d5d704c5d
vendor/mldsa-native/proofs/cbmc/polyvecl_pointwise_acc_montgomery_native/polyvecl_pointwise_acc_montgomery_native_harness.c	237	25f0386652609a7b56fc7ea3ccb52fd753de0bc0cc9d95a387554
vendor/mldsa-native/proofs/cbmc/polyvecl_uniform_gamma1/polyvecl_uniform_gamma1_harness.c	257	1c9026e4610b2c872902038aecec4f54a456f73e6d8ca0438708b
vendor/mldsa-native/proofs/cbmc/polyvecl_uniform_gamma1_serial/polyvecl_uniform_gamma1_harness.c	257	1c9026e4610b2c872902038aecec4f54a456f73e6d8ca0438708b
vendor/mldsa-native/proofs/cbmc/polyvecl_unpack_eta/polyvecl_unpack_eta_harness.c	215	e24c14259c7a72c3443d634abb5348f7e63b0af4dd6a30fac60f4
vendor/mldsa-native/proofs/cbmc/polyvecl_unpack_z/polyvecl_unpack_z_harness.c	213	80a9e70b0d04bbd374d060c4f5c4574c93b66002e20b8ce5b734b

vendor/mldsa-native/proofs/cbmc/polyw1_pack/polyw1_pack_harness.c	203	98a6533440ed3a273f45e658d54b07bd931d08a80a5ae4109cca3
vendor/mldsa-native/proofs/cbmc/polyz_pack/polyz_pack_harness.c	202	97bd63a11889029bfce54e035f821c16ff7b72333dabe42265bec
vendor/mldsa-native/proofs/cbmc/polyz_unpack/polyz_unpack_harness.c	204	167850a9aae54273f22f4f3c61774192a2ae90dcf8cc4b6e5e912
vendor/mldsa-native/proofs/cbmc/polyz_unpack_c/polyz_unpack_c_harness.c	365	8d4a68a217b1f29174eab59348bcd0f9dc21ce69d147ae8049cc0
vendor/mldsa-native/proofs/cbmc/polyz_unpack_native/polyz_unpack_native_harness.c	204	167850a9aae54273f22f4f3c61774192a2ae90dcf8cc4b6e5e912
vendor/mldsa-native/proofs/cbmc/power2round/power2round_harness.c	214	da7ca4f0fe1941cc884a9d6220e32ee3726a5fe00b0bdd41e0bad
vendor/mldsa-native/proofs/cbmc/prepare_domain_separation_prefix/prepare_domain_separation_prefix_harness.c	338	f35957138eac0b01011cc8a03a71d781141f104d472c69ded9a78
vendor/mldsa-native/proofs/cbmc/reduce32/reduce32_harness.c	197	725225bb92a0e44a4e9102a836bf05aace2beb90b069a88cb9e39
vendor/mldsa-native/proofs/cbmc/rej_eta/rej_eta_harness.c	492	5a4e8571621f0ae80b9a2da803e820648a6c0214f41ce3a86be5e
vendor/mldsa-native/proofs/cbmc/rej_eta_c/rej_eta_c_harness.c	555	984de377f74fd69703f22c3c2ecba12b02bfd86f56af05dd84a95
vendor/mldsa-native/proofs/cbmc/rej_eta_native/rej_eta_native_harness.c	492	5a4e8571621f0ae80b9a2da803e820648a6c0214f41ce3a86be5e
vendor/mldsa-native/proofs/cbmc/rej_uniform/rej_uniform_harness.c	508	978cfa69c16c19ad668eea07cb18237ad19ce1309d37a80c42d7d
vendor/mldsa-native/proofs/cbmc/rej_uniform_c/rej_uniform_c_harness.c	516	3df2cba3e53afa788b334709c77740763d52b35fc0509bd250b60
vendor/mldsa-native/proofs/cbmc/rej_uniform_native/rej_uniform_native_harness.c	508	978cfa69c16c19ad668eea07cb18237ad19ce1309d37a80c42d7d
vendor/mldsa-native/proofs/cbmc/run-cbmc-proofs.py	16123	0cf33268a521d573aaa2c2063265dd0986bbb2e7fda50c5bce259
vendor/mldsa-native/proofs/cbmc/shake128_absorb/shake128_absorb_harness.c	254	ffe929e975d1858338b860516f649bcefb3ba8a67b34202deeea6
vendor/mldsa-native/proofs/cbmc/shake128_finalize/shake128_finalize_harness.c	208	6cc48af79f24865e33690bd3207e382cc0e81ea018ff5c647d305
vendor/mldsa-native/proofs/cbmc/shake128_init/shake128_init_harness.c	204	3852905daf05da84e2278f38836323066abe5a84fafb0c7272c00
vendor/mldsa-native/proofs/cbmc/shake128_release/shake128_release_harness.c	207	c803522db52c3c1bdd76dd5875215505ffe3612ac063c71ce32ce
vendor/mldsa-native/proofs/cbmc/shake128_squeeze/shake128_squeeze_harness.c	261	a331241aa65a8ab8ab9124a145443161428e5e1848be3d2928688

vendor/mldsa-native/proofs/cbmc/shake128x4_absorb_once/shake128x4_absorb_once_harness.c	457	2673de3726cac76151641bb219f4826052bd299287610e7aa074d
vendor/mldsa-native/proofs/cbmc/shake128x4_squeezeblocks/shake128x4_squeezeblocks_harness.c	520	df05655fb74dd32d8644092575a659383477d3ad9e11f97748965
vendor/mldsa-native/proofs/cbmc/shake256/shake256_harness.c	268	f0252e6a66aebcc7f4377bdd924b1c57746f58a2f3bfa6cb690d0
vendor/mldsa-native/proofs/cbmc/shake256_absorb/shake256_absorb_harness.c	254	25d04147e7948c34b95b921222a5fd87978db34cd749748301e73
vendor/mldsa-native/proofs/cbmc/shake256_finalize/shake256_finalize_harness.c	208	7b3d3a4f37d1005377d0884d93ca6b771c9a1fc43461c5cd7c078
vendor/mldsa-native/proofs/cbmc/shake256_init/shake256_init_harness.c	204	fa067930119c65ed016589ac289e558cc55d866d276c1aab72983
vendor/mldsa-native/proofs/cbmc/shake256_release/shake256_release_harness.c	207	5fba1e02a3c7cc7589edba89bf49ad08d6226bbdbb103a0055ff2
vendor/mldsa-native/proofs/cbmc/shake256_squeeze/shake256_squeeze_harness.c	261	4c4db9c83ca163a9d1e609967aaa40418086d16919fefb2726e82
vendor/mldsa-native/proofs/cbmc/shake256x4_absorb_once/shake256x4_absorb_once_harness.c	457	5f009ba561567b35e620e0b893a547361efbf444a40c35e8e5d8d
vendor/mldsa-native/proofs/cbmc/shake256x4_squeezeblocks/shake256x4_squeezeblocks_harness.c	520	213521bcc666f500fa149c8dabec63638818050d52164e9fe0bd6
vendor/mldsa-native/proofs/cbmc/sign/sign_harness.c	385	c55578eb683a6e15aa017a2fbb8f982712ef7c225dee1e2aac3f4
vendor/mldsa-native/proofs/cbmc/sign_keypair/sign_keypair_harness.c	282	b75d18b1af095e0c6c6691228ad684b64df49a5f38ddf1d56710e
vendor/mldsa-native/proofs/cbmc/sign_keypair_internal/sign_keypair_internal_harness.c	286	2570459c0c5720f01d04b858158a80017433683860b0feb072d15
vendor/mldsa-native/proofs/cbmc/sign_open/sign_open_harness.c	396	69bad8f112936b863f41f67d96353e9835f18e6ebb557b545014a
vendor/mldsa-native/proofs/cbmc/sign_pk_from_sk/sign_pk_from_sk_harness.c	299	2295bce73a3f8983c9504e091dcf071d6731417d13aaa6131cff1
vendor/mldsa-native/proofs/cbmc/sign_signature/sign_signature_harness.c	409	f60db4865279c7ba91cb013be960acb736e11b3972281ba926e49
vendor/mldsa-native/proofs/cbmc/sign_signature_extmu/sign_signature_extmu_harness.c	332	1836b4bddf8d4fe59ef08857440ff0b07004fcfe0310d261b33d1
vendor/mldsa-native/proofs/cbmc/sign_signature_internal/sign_signature_internal_harness.c	456	9f08ad9e854e850866ea1d5e4cd3cee72e7b675a002281d2d8aa2
vendor/mldsa-native/proofs/cbmc/sign_signature_pre_hash_internal/sign_signature_pre_hash_internal_harness.c	487	062764b60b7a236b5d23cf2305069f76cb7207844cbcf901e08ff

vendor/mldsa-native/proofs/cbmc/sign_signature_pre_hash_shake256/sign_signature_pre_hash_shake256_harness.c	459	c417ccc690fd8494d8f616589b773b7f564c789ce8069b6a882ce
vendor/mldsa-native/proofs/cbmc/sign_verify/sign_verify_harness.c	427	0bb35259822af765896d6e0e5696eaae5c0eba8bad386a47fadc8
vendor/mldsa-native/proofs/cbmc/sign_verify_extmu/sign_verify_extmu_harness.c	352	bbba233035e102ecd3612f2c182d7b1480eba3595921031ae2d05
vendor/mldsa-native/proofs/cbmc/sign_verify_internal/sign_verify_internal_harness.c	457	8b2399f221589beeafc508221db1daf952dcc015c9fa9ac778739
vendor/mldsa-native/proofs/cbmc/sign_verify_pre_hash_internal/sign_verify_pre_hash_internal_harness.c	462	f8097d88c40a83648bffdaff7be4e5ae2ec16d2ae2bc3943b0ed9
vendor/mldsa-native/proofs/cbmc/sign_verify_pre_hash_shake256/sign_verify_pre_hash_shake256_harness.c	434	300e312dd72cf1a6e419351e5074d05cf7ba10467b1ee0a1eb7f9
vendor/mldsa-native/proofs/cbmc/sys_check_capability/sys_check_capability_harness.c	261	dacdec03b1cece8f747f4aa0f760a6f1a0e7e3af2c42463eae094
vendor/mldsa-native/proofs/cbmc/unpack_hints/unpack_hints_harness.c	342	ad42b5e0e26fe0fd278b32ab4df5bf130b4588419caf4dad41ded
vendor/mldsa-native/proofs/cbmc/unpack_pk/unpack_pk_harness.c	213	6c61550567df357d0c730bdc1495982a993ae8720c7c951c313a2
vendor/mldsa-native/proofs/cbmc/unpack_sig/unpack_sig_harness.c	263	3cdcbacd35378bac719449eaff2592b6ca7a6fa2fab16d16bb0eb
vendor/mldsa-native/proofs/cbmc/unpack_sk/unpack_sk_harness.c	262	7bca77f595eaa715552d94bd6f809382eba95b96dc96de0019eaa
vendor/mldsa-native/proofs/cbmc/use_hint/use_hint_harness.c	201	4d037cbcea49b6b2e190eaf2db0c7d5f994723204153574f554a3
vendor/mldsa-native/proofs/cbmc/value_barrier_i64/value_barrier_i64_harness.c	185	e892a64739e6dffdff0d178c8d99c1237195d43db1e27b038890f
vendor/mldsa-native/proofs/cbmc/value_barrier_u32/value_barrier_u32_harness.c	186	5accd53c3deaa9e64b21bb5056b3f08fb07749bbe8dc9c433ebb6
vendor/mldsa-native/proofs/cbmc/value_barrier_u8/value_barrier_u8_harness.c	257	43b718b7fd4adcd05ccd0bc0b44753aa30f096fd954bee526eafc
vendor/secp256k1-zkp/contrib/lax_der_parsing.c	3595	4da2ecdc0a3b9dbb9029d188bb0243764a3c2f2324613a688f1ac
vendor/secp256k1-zkp/contrib/lax_der_parsing.h	4230	7f3e8ba009f19a5f2759c9065af02cc8e62351100d69f8ae7e762
vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.c	5112	48352c75520eecd141936c0011601925510cdf60afd3e46e79f0a
vendor/secp256k1-zkp/contrib/lax_der_privatekey_parsing.h	4016	3420812d623c0f2f86debe0535f98eab9b929036bb2fee2de8e3b
vendor/secp256k1-zkp/include/secp256k1.h	42403	328da0a2e0433e9cdcfae1fbde894d50a17c349e366e03f3f99fe
vendor/secp256k1-zkp/include/secp256k1_ecdh.h	2560	ad314d14cb79a79d4113862f21e0deccf37962db7811c801dab28

vendor/secp256k1-zkp/include/secp256k1_ellswift.h	9148	620db468b1c13968946e9f404ceb59a3f11c895603bf1dcd33995
vendor/secp256k1-zkp/include/secp256k1_extrakeys.h	10870	7c3468c9fd192ad152c5cf042cb44938961d752fcf19966123ce6
vendor/secp256k1-zkp/include/secp256k1_musig.h	28115	b7e7772dfc78585dac1d7c2921a44dc62b3aec92df50818fdf985
vendor/secp256k1-zkp/include/secp256k1_preallocated.h	5966	b7283d4cc9c13bb2572dc92e69cea76f3270403baf5748ea4d91a
vendor/secp256k1-zkp/include/secp256k1_recovery.h	5250	e00678a1108acb36c6631a1e290cf33813b0ed5965d2322636700
vendor/secp256k1-zkp/include/secp256k1_schnorrsig.h	8177	8185d6c6ea3234fff05f38fad4b1b6f48f0574ab599dfc87297ab
vendor/secp256k1-zkp/src/assumptions.h	4667	cc875f356cc8f7f272def71f7d7261e44cc27a18a5b668cd8192c
vendor/secp256k1-zkp/src/bench.c	10186	08b5b70457175cfd2928d361ae7f0ac2c055c2de0857523ac647c
vendor/secp256k1-zkp/src/bench.h	4778	82667b60e67a152936f8b3e7be8eb0224dc8f6f806b51edd0c4c1
vendor/secp256k1-zkp/src/bench_ecmult.c	15580	c742aeab4b1af836ac736f5adc78d28482c9684a6823c01713609
vendor/secp256k1-zkp/src/bench_internal.c	17480	d7704fd54f646f295497b9f8c12c2cf82eab2f42179de64bcbaef
vendor/secp256k1-zkp/src/checkmem.h	5141	450e7928206b994693912ad5c9bbb890ca5250feded149f5025f6
vendor/secp256k1-zkp/src/ecdsa.h	1164	f97753416b0971133ad9d8acf9af02259931d0285a0333a975320
vendor/secp256k1-zkp/src/ecdsa_impl.h	10462	1e6041a11a45baa093df0c8037035bc0ac5a9a2b922cc2c9966aa
vendor/secp256k1-zkp/src/eckey.h	1449	cf511599285faecd27ace6191eca3945d85bc852b11076a1b1727
vendor/secp256k1-zkp/src/eckey_impl.h	3457	57ae9e8d50dc04bf7ee54cd13e56f1801c3cb587de6baa81c630f
vendor/secp256k1-zkp/src/ecmult.h	2679	3158078d831594da3ff1222d8547ddca1a6ca28335ded360100fd
vendor/secp256k1-zkp/src/ecmult_compute_table.h	1073	5a8ce69b18d777c3ad608ef173fd6a47c75b9c37bc6801f19ca4c
vendor/secp256k1-zkp/src/ecmult_compute_table_impl.h	1913	787c123550f8a7d03a72463abec9057ef7fa60b55ddbbb4fd274e
vendor/secp256k1-zkp/src/ecmult_const.h	1354	4705e7baabcbf8fd1853f8cef45a7f6a3075b98925d91cc209afd
vendor/secp256k1-zkp/src/ecmult_const_impl.h	19584	07238bcf17c3292bb8198f7f37baf068bdc9a6a57f843c439e220
vendor/secp256k1-zkp/src/ecmult_gen.h	5368	992ad0e8f2c0dd4cb699af8a5db7c640e904e7696b98f008d9a16
vendor/secp256k1-zkp/src/ecmult_gen_compute_table.h	672	51eb54b0cb7b49b25cf751efb2799322d73c97bc9bdd2a263feff
vendor/secp256k1-zkp/src/ecmult_gen_compute_table_impl.h	4705	9ae524743bb5871bff7e41e48b61218336b1919769132aafa9442
vendor/secp256k1-zkp/src/ecmult_gen_impl.h	16392	12e5d89186b8663e82eccbb868d9e82b266b06d9c931aa7c4cc17
vendor/secp256k1-zkp/src/ecmult_impl.h	34279	1d9505cf8d42d9c3e2006288bd95c48f876cf87701d8095fc7181
vendor/secp256k1-zkp/src/field.h	15120	481acd5c23d25ca8f5d963616ab9c3e94cf003fdb4b24608def01
vendor/secp256k1-zkp/src/field_10x26.h	2338	ba9a43b62e7e70e6e18af2787bd01dfc3a57fea527cf8ad033319
vendor/secp256k1-zkp/src/field_10x26_impl.h	52106	6f209a533fd96e73f4c92b101bd3ca46a2f7553967ba7f44650a6
vendor/secp256k1-zkp/src/field_5x52.h	2380	cbaa91c253ba84d99efcd334c7a39906c164d5edf3cde81200a44

vendor/secp256k1-zkp/src/field_5x52_impl.h	19360	8104a1b630635c9cc89628142c9fa7afc6f31a888fe50c8ebb2bd
vendor/secp256k1-zkp/src/field_5x52_int128_impl.h	11341	ee6d585731507c51dc8acd819b66556cf0cd804da1a4b6072a79c
vendor/secp256k1-zkp/src/field_impl.h	14052	33cb488ee009334a7a2ac52a7eabe39795bd184da0238e7acf73a
vendor/secp256k1-zkp/src/group.h	11798	f5f85b79a0c1659bf0a85ff199844b60fc2ace13d46c15fcac518
vendor/secp256k1-zkp/src/group_impl.h	34553	f55f65b95486c79ab44b55a767fb0879b27983a119757f9207b3c
vendor/secp256k1-zkp/src/hash.h	1916	e0e586994dd4eb8d6fc6b1846519c1f71f00cf3c2f956264fb265
vendor/secp256k1-zkp/src/hash_impl.h	13620	1dc20950aea3e66745a3f44f902e1de94b7effa8f3f82e40312f4
vendor/secp256k1-zkp/src/hsort.h	1685	220cb669835d34c252598ecff860c93edda337a9f8c2fc07b2b6c
vendor/secp256k1-zkp/src/hsort_impl.h	4798	2861489c01b927b27d99efe173799a812434748b4addd30d10218
vendor/secp256k1-zkp/src/int128.h	3622	89ac5d91d64f857038d063dd70eec4b29b4f32aa218c790fe0491
vendor/secp256k1-zkp/src/int128_impl.h	376	5683c02dfc495639da259d80624711c1c449bbaee6186729e399e
vendor/secp256k1-zkp/src/int128_native.h	581	b4c52c7ff091083f5eca9879b98641c7a38356a3c5321cbcc22ae
vendor/secp256k1-zkp/src/int128_native_impl.h	2881	ecaafefadbde3db034d4f44678096833d2d113fd21b4ff7044095
vendor/secp256k1-zkp/src/int128_struct.h	229	0733b3a65719a1e9a8fb3564c71a38f2f31ca3522f40087d536db
vendor/secp256k1-zkp/src/int128_struct_impl.h	7884	0b4e2200327f5357078ba43b0e0514c58c2b1eb217b87ebc14d95
vendor/secp256k1-zkp/src/modinv32.h	1865	774887ccd462b145434d1d6b9ae9455aa3618e791544a2b61267e
vendor/secp256k1-zkp/src/modinv32_impl.h	33492	befe02af7099a871677c7b05a81302dff5a4dc1d59f19b20bcb58
vendor/secp256k1-zkp/src/modinv64.h	1969	b71cf177d768beac28fcb4f0ef545aafaad16ad0f275c70ee4c60
vendor/secp256k1-zkp/src/modinv64_impl.h	37653	1254dc7ea941fc4d4ef0a85eed80db58d9e2647a8b8d19238d6af
vendor/secp256k1-zkp/src/modules/ecdh/bench_impl.h	1761	3929ead352b3b6425f0fdf94a1b2268d124cc0176929ae6ea3b4f
vendor/secp256k1-zkp/src/modules/ecdh/main_impl.h	2500	c507af8f5c6a408765ed96879739737a86cf0da64a65e9e5d0f6e
vendor/secp256k1-zkp/src/modules/ellswift/bench_impl.h	4399	8973d64bda27eb72a4807aae665a1a2932d624370f9d40f8833f7
vendor/secp256k1-zkp/src/modules/ellswift/main_impl.h	26216	707f86acbd1bef943bf313185bfa72068062ef385d33750e1cf46
vendor/secp256k1-zkp/src/modules/extrakeys/main_impl.h	9961	a58762afb6a494aaec984d2e65adacf1125be5e55ad2904af0f26
vendor/secp256k1-zkp/src/modules/musig/keyagg.h	1288	4d8bfe0a8955d2789205df2cd90e6bf9f53d0526f108bcbed0514
vendor/secp256k1-zkp/src/modules/musig/keyagg_impl.h	11081	d3604de574cbf96fd7c0656c6ff9500645bf27f769709928e7336
vendor/secp256k1-zkp/src/modules/musig/main_impl.h	426	9ec7e6082aa496353466769bb9f7705c32caf8d060b0c92815cb3
vendor/secp256k1-zkp/src/modules/musig/session.h	861	c256c3742a688495205698456 40f152a22a4a8c21a96040cf0c10
vendor/secp256k1-zkp/src/modules/musig/session_impl.h	30822	9dcc4c823bd1bb9c1e68d0e2a656528f5c45ab2c9d2cf3f659455

vendor/secp256k1-zkp/src/modules/musig/vectors.h	37797	c7cd9dc3804d864d17e43c5b00335a405a3bc4a30861838ac4599
vendor/secp256k1-zkp/src/modules/recovery/bench_impl.h	2320	20d13192a4aabaa52ac4c2a2da772d8aafef00773051836b25faa
vendor/secp256k1-zkp/src/modules/recovery/main_impl.h	6135	fbec2358a560b669d85720108b0b1a8f653c91531a88ded0a1da3
vendor/secp256k1-zkp/src/modules/schnorrsig/bench_impl.h	3967	1e9f3b239e1fac3f2b9f2c2e5d093f9d9811b1554e272a52f3ba8
vendor/secp256k1-zkp/src/modules/schnorrsig/main_impl.h	10281	d627be29e9a7e1e3b607df983c9f4e7b66990f056c7fcc242a16e
vendor/secp256k1-zkp/src/precompute_ecmult.c	4014	9d4d5f79ae0b7c2a2283b33486a397e2a4de40a68a35d15e3e52d
vendor/secp256k1-zkp/src/precompute_ecmult_gen.c	3879	bf49c461eda49ce4967396a3f6cf25aa92ec7029efb7609836fec
vendor/secp256k1-zkp/src/precomputed_ecmult.c	2409168	1fba19c482de09f566ed9e7bf6a526e4edb59a1cf38218e3ab413
vendor/secp256k1-zkp/src/precomputed_ecmult.h	1585	d792b8a49759deff5f1e4092558fa9ae11d24e02695e44bee63a8
vendor/secp256k1-zkp/src/precomputed_ecmult_gen.c	259715	c0a7af0884b7b07962be67e9da674e0d75a8d412d116153b8eae2
vendor/secp256k1-zkp/src/precomputed_ecmult_gen.h	973	8f514bed441bccf7e34ff05ff49d3a44e9395519a8fdcf176e6ae
vendor/secp256k1-zkp/src/scalar.h	5460	cdc9a0053321cc541c81bb02f5c4dbba1b9489947908b3187a7c3
vendor/secp256k1-zkp/src/scalar_4x64.h	801	42aed583b28bda7982c54a6f63e50c0ae5f9c27d5f66d92ec6bf6
vendor/secp256k1-zkp/src/scalar_4x64_impl.h	32955	992b203310337ad913afbdc29691f184462aba37d121c13a76581
vendor/secp256k1-zkp/src/scalar_8x32.h	725	0614cea113aea88a47ef3b2ba8167722e850d7b2a696a4f16b03e
vendor/secp256k1-zkp/src/scalar_8x32_impl.h	29162	7dd402b15ae1c9b884f20a86c952db4aabb7c13fdba0a4d9addc7
vendor/secp256k1-zkp/src/scalar_impl.h	12113	6ba654bd4edc0349d1d7f1dfc25ba2d73dd9f2ecdca8078e384bf
vendor/secp256k1-zkp/src/scalar_low.h	1135	3cab20db10224ff85ee7c6d975d2968c1370ed8e96ea26b94650f
vendor/secp256k1-zkp/src/scalar_low_impl.h	5967	b44c726ac42516345dff05b1d964807609e7c53112aefe584aa0a
vendor/secp256k1-zkp/src/scratch.h	2224	4af5e1ff665c0418bcc5b1c2de954e7b377f9bdfdd4309e029693
vendor/secp256k1-zkp/src/scratch_impl.h	3677	294a55d429fa4cc9dec46ade345bbb2bfdff4055eb1cbf1ba6168
vendor/secp256k1-zkp/src/secp256k1.c	29096	6a3cad3929bfd49c8f99d36c3e949f4067380dd40c4ba6b81a1dd
vendor/secp256k1-zkp/src/selftest.h	1271	fa95cb3e35c5d121da797653a58c4c43370d51245aa43b65e0292
vendor/secp256k1-zkp/src/testrand.h	1886	b1708cdab67d44311b97af6f77c719f6dfc03d8c53e5e686bc1d9
vendor/secp256k1-zkp/src/testrand_impl.h	5585	5410f491bb7dc8e40715b597fb22c2fe16df52602beea32a2e045
vendor/secp256k1-zkp/src/testutil.h	4355	c904296216746f591f0404ef9507a93090bdd5b2826a28834154d
vendor/secp256k1-zkp/src/util.h	15742	a2d678ac4126c742cee549dda6de8b2def62cbd87103ec7e9fd42
vendor/secp256k1-zkp/src/util_local_visibility.h	402	d47d4a58f40157ab2a298151f366c5ebec8f7a1e7aaba88e36b53
vendor/secp256k1-zkp/tools/symbol-check.py	2416	1706069df728c1a2ac1d2578897076713c16eed73315be6c83ad9

vendor/secp256k1-zkp/tools/wycheproof_utils.py	427	9f2c2368953eeb3c9b7c3b31a807cde76b3106df93fbb97599363
vendor/sphincsplus/benchmark.py	1333	6925bba99222e3667eef0e54416df34748650c868737e4143ee97
vendor/sphincsplus/haraka-aesni/context.h	224	9bc8fdb0a0806946f5323c2c6eceb380380dbd0f84b23321ef3f1
vendor/sphincsplus/haraka-aesni/haraka.c	22888	1bbec250d990bc84459f12bcdcfae1c82d2ab1f6eba55daa8c541
vendor/sphincsplus/haraka-aesni/harakax4.h	1165	68a1c2f8f83f5f89ffb5fc4b02b1a81cd4a4da4e5cc9880a4d908
vendor/sphincsplus/haraka-aesni/hash_harakax4.c	999	e0ee5b5b6b44ae8f04038eee5223e8638b57954e84b41c04e4012
vendor/sphincsplus/haraka-aesni/thash_haraka_robustx4.c	3506	426efc721b11778b64e84216280c4854d5c19fe164340b1b5df7e
vendor/sphincsplus/haraka-aesni/thash_haraka_simplex4.c	2307	0e83c32e43788e7a364db2c517c87d2511ce6b2019d37afdf5e3c
vendor/sphincsplus/ref/PQCgenKAT_sign.c	7001	793da075036e906843d0ef33fd28d39761a08ee94cd84c98c0cef
vendor/sphincsplus/ref/address.c	2957	e07ecc87e564db0f2055b761d9b9a56edec7cb7508e3410117e85
vendor/sphincsplus/ref/address.h	1683	86b33648b11e72d09c21e4b478881f5e6aaf6770ef47a29df6fd7
vendor/sphincsplus/ref/api.h	2116	a0c7abd648a4a695ff031f034a68e431559338c37890b428f21ce
vendor/sphincsplus/ref/context.h	479	6953afd8e8d45fab9f47bf8beeac549fdb45aeb02c2d2e263c130
vendor/sphincsplus/ref/fips202.c	19100	f2a076880fa2d32e585c4ba30750171b4b842591d1f6ade6038cf
vendor/sphincsplus/ref/fips202.h	1676	125d3ac43b0c81b4b796de97955d8016478183fd4c10feb73858e
vendor/sphincsplus/ref/fors.c	5322	efa5c96c53814d84ea9a142dcf966adf17873a0f73a7a8704c242
vendor/sphincsplus/ref/fors.h	1104	349cbe538ae0c6510b8ceeb9685e40b226a7386ce2e2454c3269c
vendor/sphincsplus/ref/haraka.c	26095	529c127e606f47982dc3ac46b43b40d295098bddb2cbd93d703d9
vendor/sphincsplus/ref/haraka.h	1494	b48a0c762064217119557cc2e432ebcef2b063981a6fbd4f5b1df
vendor/sphincsplus/ref/haraka_offsets.h	1059	499ba06ce128175145c661021a23cf140bba2e95288c664a5352d
vendor/sphincsplus/ref/hash.h	967	2b09886addb6b9628e645d0054bc06ff1eb418f59a778c8a4b741
vendor/sphincsplus/ref/hash_haraka.c	3032	e8919ed13da9675288d12b333dc50ea0ba3706d13ee8e618792d4
vendor/sphincsplus/ref/hash_sha2.c	6714	bc804b4d9b4df95531a8304dfdc01b7438870672d30ab0e7ca34a
vendor/sphincsplus/ref/hash_shake.c	3063	b4720bbf56f843e333391580ad9e9437f464f502331d3ce06e114
vendor/sphincsplus/ref/merkle.c	1973	822c402a9dbe43274d2dad92ff215e80b20a5464c16ac59ee57d2
vendor/sphincsplus/ref/merkle.h	571	2f704262633f56514218bb79798079102e50393879520c9d05b2b
vendor/sphincsplus/ref/params/params-sphincs-haraka-128f.h	2131	8d67a228f57719cab9f8bc581af5fc321abb0393167adb1d95a78
vendor/sphincsplus/ref/params/params-sphincs-haraka-128s.h	2131	bf9ea2f8825d64f14ee2f87504394b84e7d9c831080b26899123e
vendor/sphincsplus/ref/params/params-sphincs-haraka-192f.h	2131	b127c87293548adf91d4e02cbb96968b3fa6a21f2fea02193d00c

vendor/sphincsplus/ref/params/params-sphincs-haraka-192s.h	2131	0cf3601315cdc13ca78a6ae85ddc7b00e61363fe17eca9387afb5
vendor/sphincsplus/ref/params/params-sphincs-haraka-256f.h	2131	bae1adff2781a5a9aad0c20b9d11b20cc5518a858449555227a37
vendor/sphincsplus/ref/params/params-sphincs-haraka-256s.h	2131	6c329295f338557ed43495eee00a31bcacb8c948e3ce9a3ea344f
vendor/sphincsplus/ref/params/params-sphincs-sha2-128f.h	2348	4bb379eec38814fd6a817016bc62810ad055e38f8902d268a464c
vendor/sphincsplus/ref/params/params-sphincs-sha2-128s.h	2348	58df8304a35570699ac875b2cf233baee9127ec4df29f2a546125
vendor/sphincsplus/ref/params/params-sphincs-sha2-192f.h	2355	570eed62eae97d14385f678965ce293bb51030d4be1a216a58663
vendor/sphincsplus/ref/params/params-sphincs-sha2-192s.h	2355	d14c7c6a5754f48a583453a693a36bb293dd70b1733bd3ccd4c7e
vendor/sphincsplus/ref/params/params-sphincs-sha2-256f.h	2355	59ad3656e20d674329b69ce972e4f3e1cfc8778f73219875b2cd7
vendor/sphincsplus/ref/params/params-sphincs-sha2-256s.h	2355	6648b7c2acfeb1aaa410ee9038235175ec64679db2c371cd3cde3
vendor/sphincsplus/ref/params/params-sphincs-shake-128f.h	2130	de37a1ad4de21af24c70262a97b74927a0decb8275eae1ad34dcf
vendor/sphincsplus/ref/params/params-sphincs-shake-128s.h	2130	da4f8c1ee5b475a8103d856d80dbec15276eb2f566ce44318dcd2
vendor/sphincsplus/ref/params/params-sphincs-shake-192f.h	2130	5524ce6e92393fa176d3f3897c612a402386c8ae1a2f089f53b4e
vendor/sphincsplus/ref/params/params-sphincs-shake-192s.h	2130	75fae04788d85078b605b540a45d8478251c2c8bd0b052972e1b3
vendor/sphincsplus/ref/params/params-sphincs-shake-256f.h	2130	5042072dde059ebafe3b938b8e31e3c3ca97dbc70a7a01407323e
vendor/sphincsplus/ref/params/params-sphincs-shake-256s.h	2130	11852f080cb720d08026d2dda6fe410a5cba933dca5b70332f9e8
vendor/sphincsplus/ref/params.h	81	a413a8ac8f601b4631980cb1b11201def25591e9423254301ee6f
vendor/sphincsplus/ref/randombytes.c	762	5cfaf24e8a2d130b2fec5def511f635c18d381712e78050124eca
vendor/sphincsplus/ref/randombytes.h	129	cb0022583e425fcf2cbb9227519893e336d194b04e7d8083c4ed6
vendor/sphincsplus/ref/rng.c	5363	3f7fff913ed542ab52a29c0ce4498b7a271f8d4d6abc6d038bab5
vendor/sphincsplus/ref/rng.h	1216	0c438e7e18515030d902540060cf6755486b17b43d1f1f0b1a790
vendor/sphincsplus/ref/sha2.c	21257	cbe87248a6a794cf92df33759cbad5ac5f1a2df390edcbb292f07
vendor/sphincsplus/ref/sha2.h	1368	c5b7768e75d52c68111b3cdf3e1b192d52bccb85e0ba9fa0ac243
vendor/sphincsplus/ref/sha2_offsets.h	1042	fbbece33219c2fcc4c76616ea779fbec1a977715a9fc0ac2c9c1b
vendor/sphincsplus/ref/shake_offsets.h	1055	8aba4f437e28c23e0c020fbf9d68422718bdd15544c46d1fe6e4f

vendor/sphincsplus/ref/sign.c	8147	4015a353fa2b0433de4c1d0496a847cede14ffd7f9e60c92a25c8
vendor/sphincsplus/ref/thash.h	276	9a81d94591cb7d97606b0927525e59994b70b0043301f47861f98
vendor/sphincsplus/ref/thash_haraka_robust.c	1324	ec2068778c4ba6a75be14cdf959a185127dcea5fedecea6f09c6a
vendor/sphincsplus/ref/thash_haraka_simple.c	1020	42aa839c7d1b0a75cbd0e77fba93ff31a8438bfea891fc3a8222b
vendor/sphincsplus/ref/thash_sha2_robust.c	2376	794697e3b37e6031d209ddb0da003d0154c8739ca8cd48f20e11b
vendor/sphincsplus/ref/thash_sha2_simple.c	1793	a6da924ab60437c37466a1fd2afe65cf9d7ae8edaba8458d34520
vendor/sphincsplus/ref/thash_shake_robust.c	831	7e0123842233891387 48ffeadd2bcdcabf6693d2d69fdb20d4719
vendor/sphincsplus/ref/thash_shake_simple.c	641	a15e1c60a9fc3f4aec37abb95c117c360b1f91e93a08dc0a00953
vendor/sphincsplus/ref/utils.c	5332	9c3d1b5929997d3986d66c826a73c0f2cd39b36c68122c8823719
vendor/sphincsplus/ref/utils.h	2378	06d3eaf20944104873d9e143dce2c049719d4aff4a5073c70551b
vendor/sphincsplus/ref/utilsx1.c	3787	c66cb53481cc2a02c44e53e5ec428d9565b51ee3554e1445d29ae
vendor/sphincsplus/ref/utilsx1.h	1009	1df02869660b96a0e79205bfa6bf09b999392c42aeb5469f64b56
vendor/sphincsplus/ref/wots.c	3225	e0624601c3bfa0c2878ed7be1efb75a490a10453d3e2ca7091088
vendor/sphincsplus/ref/wots.h	663	18a7c81e8006bef01e33e2894f51f3d54583f43c29a631691cbb1
vendor/sphincsplus/ref/wotsx1.c	2269	37159c12b101033de2a53fe4125f9c5d8f254115e9e19f6b34d3b
vendor/sphincsplus/ref/wotsx1.h	1190	fdf5431c896cba8814fb5a682df259ba5267d0643e726469801ee
vendor/sphincsplus/sha2-avx2/context.h	267	63af0bb93c6c7b6647c096187fffa93ae85f420f7b1bbc0930d86
vendor/sphincsplus/sha2-avx2/fors.c	8090	353c87a66fffa6f359d4f9d2a31dd15e70555d3867f54c3bab2e0
vendor/sphincsplus/sha2-avx2/hash_sha2x8.c	2538	5704865283ff2d1c1878386fd36e5b8ca330b3d7aa2406c1f39ff
vendor/sphincsplus/sha2-avx2/hashx8.h	514	5c9cdaf7e2661649fa38f80325f9de8b46de43109f72e06d38c31
vendor/sphincsplus/sha2-avx2/merkle.c	2115	19ba62b5739fe9826e515bb5fefb765033fac3004065da9f1725b
vendor/sphincsplus/sha2-avx2/sha256avx.c	15421	dd4b332dc080704c3780613c2787f1ed6c3e1d7a639b8359d327f
vendor/sphincsplus/sha2-avx2/sha256avx.h	3512	8dbc3fd1999628825d59a928dc5227c7c8cd651c9a86186d64701
vendor/sphincsplus/sha2-avx2/sha256x8.c	7040	8089a3db6ea8006723f2610bab819b493da9ef32171788e52eab3
vendor/sphincsplus/sha2-avx2/sha256x8.h	2465	de07bcfae3e394523be4a0a3a85c263bf458f34d354f76b3d589d
vendor/sphincsplus/sha2-avx2/sha512x4.c	15546	375663933e3d89c7d47ae0c13a01fb3468f1e5bae8ecd21b56e88
vendor/sphincsplus/sha2-avx2/sha512x4.h	1137	84329a9a73a6207319897a1a44bab78a2cf77c38d14d370713f54
vendor/sphincsplus/sha2-avx2/thash_sha2_robustx8.c	11408	fab3a98115416a2eea2d82e47e1dbbf3bf2ab412d6cc9a9b4bd56
vendor/sphincsplus/sha2-avx2/thash_sha2_simplex8.c	8176	a4a3c7aa96dad3669f98c683e1daae9614fd7741237aa3ef2799c
vendor/sphincsplus/sha2-avx2/thashx8.h	817	731b861507fd7dc9dea1d77a6a1fff4f406a4e84d02ca4078e98a
vendor/sphincsplus/sha2-avx2/utilsx8.c	6238	b6a3ff08f4d0029bf4ac2741581ac2f619e2accb13940e43fc75b
vendor/sphincsplus/sha2-avx2/utilsx8.h	1084	55d8fa8004c3a2457f8a52ab8b7f0a5c59874cbaba99718edbb6b
vendor/sphincsplus/sha2-avx2/wots.c	9351	0d4fcb08b4f1927233d1b9542ac98b7db0909b93fd05ae589eb61

vendor/sphincsplus/sha2-avx2/wotsx8.h	1307	5b59d3e8e6eb9a66a598c91746c1ee4216aef974e3d99449ff351
vendor/sphincsplus/shake-a64/context.h	180	69f3f1756e1ac7cbc9827e8a83aab74d1086e9f70f505c39e5687
vendor/sphincsplus/shake-a64/f1600x2.h	212	55a55ff09c9e06979dadba873be952b9cfa17b1289449564dfeef
vendor/sphincsplus/shake-a64/f1600x2_const.c	557	3b3fde30ca2c99c95a0ff283a226999f97908fcae4fb07a84d19f
vendor/sphincsplus/shake-a64/fips202x2.c	3478	969e7567bd9dbbd2a54cb0236cdcaf83250a38a2e11e886bdf066
vendor/sphincsplus/shake-a64/fips202x2.h	624	052b5f75cf065183f4f9da0a9aeaac6f8421fa6763e50ab986989
vendor/sphincsplus/shake-a64/fors.c	6478	4d8e3d02d8bf84b9f5c00beebcbb2e3ca35860bc148f1c4fe6b68
vendor/sphincsplus/shake-a64/hash_shakex2.c	1470	743694ba9c55e60a09d4d1d7717aecb497f4629ff3934f352b957
vendor/sphincsplus/shake-a64/hashx2.h	313	6d8ebe3b64a907157b6c4f724bbb0f199ddc39f4c2e8184b4b870
vendor/sphincsplus/shake-a64/merkle.c	2027	00b3f44547c02321bee94f860610e9b49a3b28143dd14fa29e2f4
vendor/sphincsplus/shake-a64/thash.h	236	1a5ccef8bd200dc96ef9bd0d48715d32e32d7efc009c3a372f44a
vendor/sphincsplus/shake-a64/thash_shake_robustx2.c	4069	bd86c6382f5658fca4fe1920618157afde31068130890b6bd91cd
vendor/sphincsplus/shake-a64/thash_shake_simplex2.c	2815	651fb558e944c6b8c2ab4fd352450e2aa6226b9d067ab784cf540
vendor/sphincsplus/shake-a64/thashx2.h	392	4ad051f46e956ce0a3c648f4758d2e14893c06883ac5fd814118b
vendor/sphincsplus/shake-a64/utilsx2.c	5458	167806d1a38e34049bb67e7879a656336a11d188ac18a57bf7ef7
vendor/sphincsplus/shake-a64/utilsx2.h	1085	637d21c0996c970a8e227de0746c716ff812a58a2fc321b3358a8
vendor/sphincsplus/shake-a64/wots.c	8017	f30ebe58a2205e7974415949288f284824ec5fdad0a82d27757aa
vendor/sphincsplus/shake-a64/wotsx2.h	1307	19a5bff0c7c5365049aaf8d3e1383b58659a4c2b666202b155974
vendor/sphincsplus/shake-avx2/context.h	180	69f3f1756e1ac7cbc9827e8a83aab74d1086e9f70f505c39e5687
vendor/sphincsplus/shake-avx2/fips202x4.c	5440	ee5b05ad7d7d0a18707b5fec2b55e50ac4d9ed1e090fa80bc89e4
vendor/sphincsplus/shake-avx2/fips202x4.h	774	57bedbbd125434c43982b95df9e8af0dfb55f853489834366221f
vendor/sphincsplus/shake-avx2/fors.c	6998	6198284a87c362e3dc273773dc9aa0bce6f20cbd809b35cb92174
vendor/sphincsplus/shake-avx2/hash_shakex4.c	1981	c7e9d49cc93e323b70e7227cf60658a2607c179944b01ef61477c
vendor/sphincsplus/shake-avx2/hashx4.h	387	c6223dcf0c544a31be67b602015dee3fe3ab8661a14454f6c91b1
vendor/sphincsplus/shake-avx2/keccak4x/KeccakP-1600-times4-SIMD256.c	40006	a29188e05c33050fd703657ca3f29f9db0478941e4d4864b4a3ca
vendor/sphincsplus/shake-avx2/keccak4x/KeccakP-1600-times4-SnP.h	3114	82b498f8e116b736cd77e1d1665300aa75606b354e02a90cb764c
vendor/sphincsplus/shake-avx2/keccak4x/SIMD256-config.h	150	2118d0bf6aa447bae934789610f2b1b61c29ed28702d470a64dc6
vendor/sphincsplus/shake-avx2/keccak4x/align.h	939	5033313cf7203ce3514cc22d8ce113b430237e8b15a6c66a79d3e
vendor/sphincsplus/shake-avx2/keccak4x/brg_endian.h	5546	5fbc8e476a28827e0fb764db47b4e94947fd9138cd9f26fce114c

vendor/sphincsplus/shake-avx2/merkle.c	2027	66fdf83d6fa3b316bc6978bbb0fadcfa748b60076f8f67bdd365e
vendor/sphincsplus/shake-avx2/thash_shake_robustx4.c	5199	456e9ef43296fd0bd11872d386545c685ed6f55407a26f3ef54b9
vendor/sphincsplus/shake-avx2/thash_shake_simplex4.c	3746	3891e608296f0b0f4c83d0904d9ca990dd9ccb60229ab445fba3a
vendor/sphincsplus/shake-avx2/thashx4.h	525	6859c5e1dcdb090d7c3127ea0dc9289b9e0d75b061083c8dc08b1
vendor/sphincsplus/shake-avx2/utilsx4.c	5872	7683265c603aa71ff4b2effb5774d370b41dfbdc3880e289d4372
vendor/sphincsplus/shake-avx2/utilsx4.h	1084	79ab80de197ce967266920d85779c94c42b38ceaf53be261ff023
vendor/sphincsplus/shake-avx2/wots.c	8464	ff111f14558268b4a9fc165fa60dea3110e5ef4b5865387062271
vendor/sphincsplus/shake-avx2/wotsx4.h	1306	97fa5f08ccfdabe4a249e24e36ecd49ff1997a91423d9bbe29a5d
vendor/sphincsplus/vectors.py	3270	799e1ead884a113139bdb169f6a589d2eafd8b9920cf5d2cf1167